A cratonic Middle Jurassic paleopole: Callovian-Oxfordian stillstand (J-2 cusp), rotation of the Colorado Plateau, and Jurassic North American apparent polar wander

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[1] Paleomagnetic results have been obtained from two sequences of the upper Middle Jurassic (upper Callovian) Summerville Formation which are located on the North American craton (north central New Mexico). The thicker of the two sections (Trujillo) exhibits a well-defined dual-polarity remanence, forming a magnetostratigraphy of at least 12 relatively short polarity intervals. The high reversal frequency is consistent with that recorded by the oceanic crust at this time. The second section (Romeroville) is thin, gypsiferous, and poorly indurated; consequently, only a short interval (5 m) was successfully sampled. The characteristic magnetization was wholly normal polarity. In the rapid reversals of the thicker section, that thickness of normal polarity is unique and thus forms the basis of magnetostratigraphic correlation between the two sections. A cratonic late Middle Jurassic paleomagnetic pole at 148.2°E, 57.3°N (k = 30.3, α_{95} = 6.0°) was obtained from the larger Trujillo sequence. Despite large confidence limits, the small sample population of the Romeroville sequence displayed the same pole. Throughout the western United States, the Summerville Formation is overlain with little or no discordance by the Morrison Formation, although the two stratigraphic units are commonly reported to be separated by an unconformity, designated J-5. The New Mexico paleopoles of the Summerville and overlying lower Morrison Formations are statistically identical at the 95% confidence level; a similar situation exists between Summerville and lower Morrison paleopoles of the Colorado Plateau. The presence of the same geomagnetic field direction in both the Summerville and the lower Morrison Formations suggests that the J-5 unconformity which separates them may encompass little time, perhaps representing only a minor break in deposition. The two newly investigated Summerville sites are located east of the Rio Grande Rift, hence representative of the North American craton; their paleopoles are displaced counterclockwise relative to those of the Colorado Plateau, and in amounts consistent with other coeval craton-Colorado Plateau pole pairs. Together, six coeval paleopoles indicate that the Colorado Plateau has rotated clockwise by about 9.0 ± 3° since Late Pennsylvanian time. The cratonic Summerville paleopole also removes the uncertainty over the Jurassic North American apparent polar wander path. The statistically identical Summerville and lower Morrison paleopoles form a substantial data set, five paleopoles, which indicate that NA APW traced a path roughly along 60° latitude. The five paleopoles stand in stark contrast to the paleopole from the slightly older (5–8 Ma) Moat Volcanics. Spreading rates in the central Atlantic do not allow NA plate motion that would include both paleopoles positions. Because the weight of the data, five paleopoles from widely spaced localities and different tectonic blocks, the Moat Volcanics pole cannot be representative of North America in the Middle Jurassic. *INDEX TERMS:* 1525 Geomagnetism and Paleomagnetism: Paleomagnetism applied to tectonics (regional, global); 8110 Tectonophysics: Continental tectonics—general (0905); 9350 Information Related to Geographic Region: North America; 9609 Information Related to Geologic Time: Mesozoic; *KEYWORDS:* Middle Jurassic, Colorado Plateau rotation, Summerville Formation, magnetostratigraphy, J-2 cusp. **Citation:** Steiner, M. B., A cratonic Middle Jurassic paleopole: Callovian-Oxfordian stillstand (J-2 cusp), rotation of the Colorado Plateau, and Jurassic North American apparent polar wander, *Tectonics*, *22*(3), 1020, doi:10.1029/2001TC001284, 2003.

1. Introduction

[2] A large amount of the Mesozoic and Paleozoic paleomagnetic data contributing to the North American apparent polar wander curve come from the Colorado Plateau, which has been shown to have been rotated clockwise several times in its history. The excellent Plateau exposures have allowed an abundance of paleomagnetic data to be gathered; therefore assessment of the amount of rotation is highly desirable in order to correct these data for inclusion into the North American paleomagnetic database. The large amount of North American apparent polar wander (APW) in the Mesozoic makes it desirable to obtain all possible data of comparable age from the North American craton for comparison. To that end, the Middle Jurassic Summerville Formation, time-equivalent to the same formation on the

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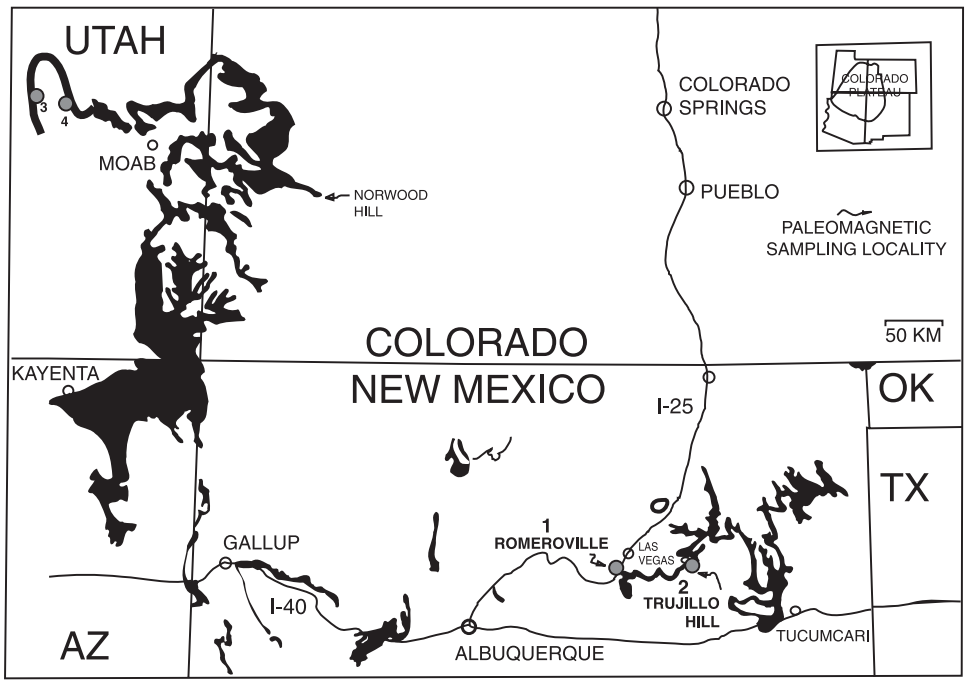


Figure 1. Outcrop map of the Summerville Formation (modified from *Anderson and Lucas* [1992]), showing the locations of all paleomagnetically studied sections (shaded circles). 1 and 2 are the localities of this study; 3 is the locality of *Steiner* [1978] and of *Bazard and Butler* [1992]; 4 is the locality of *Steiner and Helsley* [1972].

Colorado Plateau, was sampled for paleomagnetic study at two localities in north central New Mexico (Figure 1), east of the Rio Grande Rift, therefore on the North American craton.

[3] Beds of reddish-brown, evenly bedded siltstones and claystones, formerly called the Bell Ranch Formation, underlie the basal channel sandstones of the Morrison Formation in north central New Mexico. The beds occupy the same stratigraphic position and have the same lithologic aspect as the type Summerville Formation of Utah. Extensive examination of mid-Jurassic strata between the type section in central Utah and the strata of north central New Mexico led *Anderson and Lucas* [1992] to conclude that the Bell Ranch Formation of New Mexico is indeed the eastward extension of the type Summerville Formation. The paleomagnetic data reported here wholly support that conclusion.

[4] The Summerville Formation is approximately late Callovian in age [*Imlay*, 1980]. Although unfossiliferous, its age is fairly well constrained by its relationship to directly over- and underlying fossiliferous formations. In Utah, the Summerville intertongues with the underlying marine Curtis Formation which has been correlated to the fossiliferous mid-Callovian portion of the Sundance Formation. In Wyoming and Utah the Summerville is conformably and unconformably overlain by the fossiliferous late (and possibly early) Oxfordian Morrison Formation [*Imlay*, 1980].

[5] The Morrison Formation overlies the Summerville at the two localities sampled; the Morrison strata of these same localities were sampled previously [*Steiner et al.*, 1994; *Steiner*, 1998]. One locality is an Interstate-25 roadcut near the Romeroville exit at Las Vegas, NM (35.5 N, 104.7 W). The Summerville and overlying lower Morrison Formations are exposed in one roadcut [*Lucas et al.*, 1995]; the adjoining cut exposes the upper Morrison and overlying Cretaceous formations. The beds repose at an angle of 35°, dipping to the ENE. The Summerville Formation at this locality consists of 10 m of soft, pale reddish-brown, gypsiferous siltstone and claystone, resting on limestones of the middle Callovian Todilto Formation; the Summerville strata are overlain by a 56 m of thick-bedded white channel sandstones with abundant heavy mineral laminae. This association of a thick sequence of white channel sandstones containing abundant heavy mineral laminae overlying reddish-brown mudstones is notably similar to outcrops of the lower Morrison and Summerville Formations of western Colorado [*Steiner and Helsley*, 1975; *Steiner*, 1983].

[6] Samples were collected with a portable gasoline powered coring engine in magnetostratigraphic sequences. Three to four samples were taken per meter of section. At the Romeroville section, the lower 5 m of the 10 m thickness of the Summerville Formation were exceptionally poorly indurated because of a high gypsum content. The upper 5 m yielded 18 samples; three samples also were cored in the limited exposure of the underlying Todilto limestone bed.

[7] About 50 km to the east, just south of Trujillo, New Mexico (35.5 N, 104.2 W), a series of road cuts into informally named Trujillo hill, nearly continuously exposes the stratigraphic section from the Middle Jurassic Entrada and Summerville Formations, through the Upper Jurassic Morrison Formation and into the Lower Cretaceous Mesa Rica Formation. Resting on the Entrada sandstones with apparent conformity, the Summerville Formation consists of 20 m of evenly bedded reddish-brown siltstones. The

Summerville strata are overlain by 4 m of thick-bedded white channel sandstones with heavy mineral layers, the lower Morrison Formation, and in turn overlain by 82 m of variegated mudstones of the upper Morrison Formation. A total of 68 samples were obtained in the Summerville Formation, sampled from the contact with the Entrada Formation up to the basal Morrison sandstones. These beds are nearly horizontal, dipping 5° to the northeast.

2. Paleomagnetic Data

[8] The initially measured natural remanent magnetization directions (NRM_0) of the Trujillo hill samples showed a portion clustered near recent geomagnetic field directions (including both the direction of the presently observed field and that of a geocentric axial dipole); others displayed the expected mid-Jurassic reversed and normal directions or lay on great circles connecting recent magnetization directions with the Jurassic directions. NRM_0 intensities averaged about 3×10^{-3} A/m. NRM_0 directions of the tilted Romeroville strata cluster around a northwesterly direction (declination = 336°, inclination = 44°, 18 samples, $\alpha_{95} = 13°$), therefore, not a reflection of recent magnetization; intensities were slightly greater than those of Trujillo hill, averaging about 4×10^{-3} A/m.

[9] A suite of 25 samples distributed stratigraphically through both Summerville sections was thermally demagnetized in 18 temperature increments between 150°C to 675°C. From the resulting data, the rest of the samples were thermally demagnetized in 10–12 increments each, beginning at 300°C and extending through 670°C. About two thirds of the samples at both sites exhibit a direction near the axial dipole field direction as the first vector removed; that magnetization is removed by 150°–200°C. Above 300°–400°C, a number of samples display linear trends to the origin, defining the characteristic directions for those samples; many of the samples display their characteristic directions up to temperatures of 655° to 670°C. Demagnetization responses of all samples were investigated by principal component analysis [*Kirschvink*, 1980].

[10] The Trujillo hill stratigraphic sequence exhibits both normal and reversed polarity characteristic remanences. Demagnetization responses varied and can be described as four types. About one third (29%) of both normal and reversed polarity samples (20 samples) displayed linear demagnetization behavior above 300°–400°C, with lines that trended to the origin of orthogonal axes vector plots (Figures 2a and 2b). The MAD (maximum angular deviation) values range from 0.5° to 4.5°; these samples are designated as "Group I." Another 26% of the samples (18 samples) comprise a "Group II"; samples in this category exhibit overlapping unblocking temperature spectra which result in greater MAD values (5°–9°); demagnetization paths may (Figure 3a) or may not trend toward the origin (Figure 3b). Despite this degradation of the quality of remanence definition, these samples exhibit dual polarity characteristic directions which are similar to those of Group I samples. A third type of behavior is observed in one third of the samples; this "Group III" demagnetization response

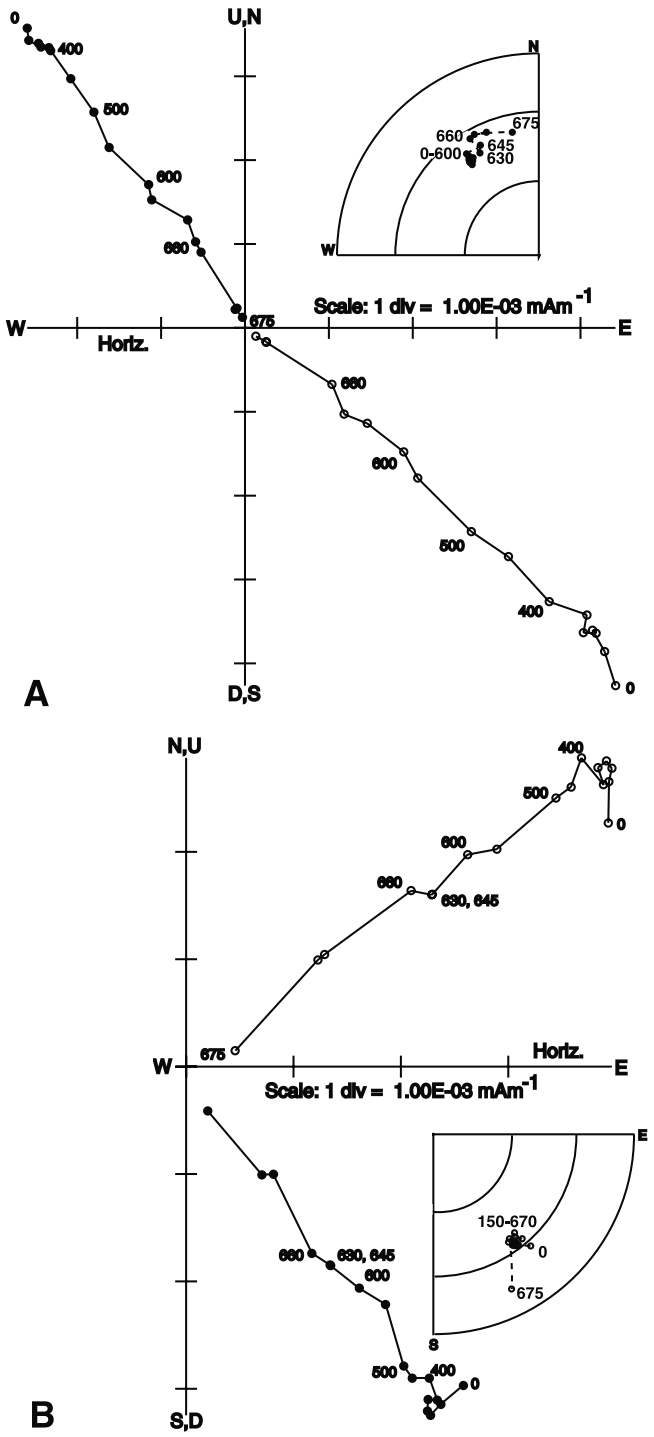


Figure 2. Examples of thermal demagnetization at the Trujillo locality exhibiting linear-to-origin behavior in normal (a) and reversed (b) polarity samples, representing 30% of the population (Group I, see text). Equal-area stereographic plots show lower (upper) hemisphere directions as solid (open) circles. Orthogonal axes plots show the horizontal projection of the magnetic vector (declination) as solid symbols and the vertical projection (inclination) as open symbols.

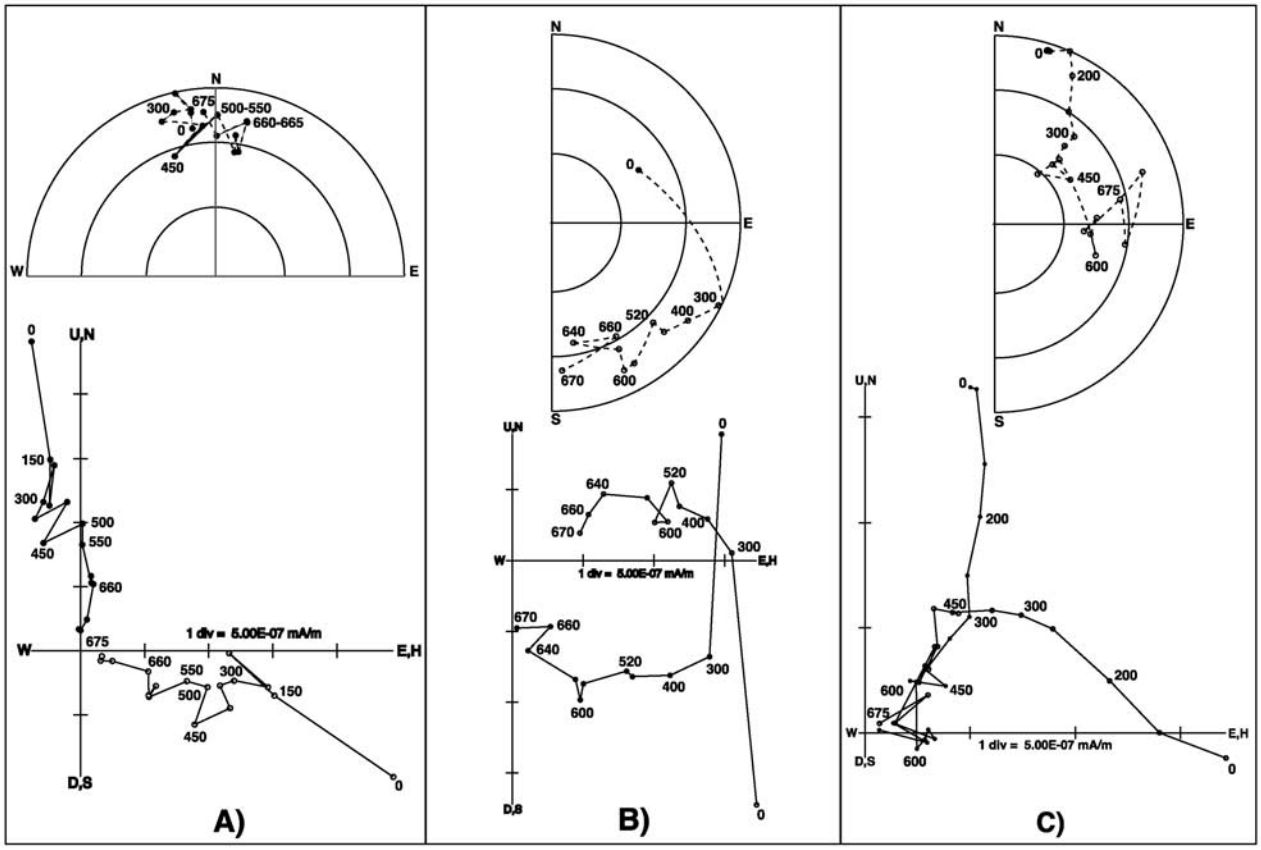


Figure 3. Multicomponent demagnetization behavior at the Trujillo locality. Typical behavior observed in 30% of the samples (Group II category) of normal (a) and reversed (b) polarity characteristic magnetizations. (c) Nonlinear behavior observed in 15% of the sample population (Group III category). (d) Multicomponent magnetization from which no single component could be isolated (Group IV). (e) Magnetizations removed during demagnetization of the sample shown in (d) suggest that Group IV samples contain both normal and reversed polarity mid-Jurassic directions.

consists of nonlinear demagnetization trends which define great circles that include either the normal or reversed direction of Group I samples (Figure 3c). Magnetizations removed during demagnetization displayed normal and reversed polarity directions similar to those of Groups I and II, and commonly directions of both polarities were observed in the same sample (Figure 3d). Finally, seven samples (10%; Group IV) exhibited entirely overlapping unblocking temperature spectra (Figure 3e).

[11] The Romeroville Summerville samples displayed demagnetization behavior similar to those from Trujillo hill, although far more of the Romeroville samples exhibited significant overlap of unblocking temperature spectra. Many of the samples from this locality would be classed with Trujillo hill Group III samples. Five of the samples were too gypsiferous and did not survive thermal demagnetization. Only normal polarity remanence was exhibited by the samples in this stratigraphic section.

[12] The samples taken from the Todilto Formation limestones immediately underlying the Summerville Formation were extremely weakly magnetized; thermal demagnetization to 400°C reduced intensities to below the noise-limit of the 2G cryogenic magnetometer ($<5 \times 10^{-6}$ A/m). Neither clear normal or reversed Jurassic magnetization directions were obtained; the Todilto limestones at this locality simply are too poorly magnetized to yield useful information.

3. Results

3.1. Summerville Magnetostratigraphy

[13] The characteristic magnetizations of Groups I-III samples are shown in stratigraphic sequence in Figures 4a and 4b. The Trujillo sequence indicates numerous changes of magnetic polarity (Figure 4a). The dominance of reversed polarity interspersed with short normal polarity intervals is the same as was observed in an earlier magnetostratigraphic study of the Summerville Formation in central Utah [*Steiner*, 1978]. The 5 m of the Romeroville section display only normal polarity (Figure 4b). The thinness of both stratigraphic sequences is a reflection of their basin-edge locations within the Summerville depositional area. The thickness of the two differs by a factor of 2, even though

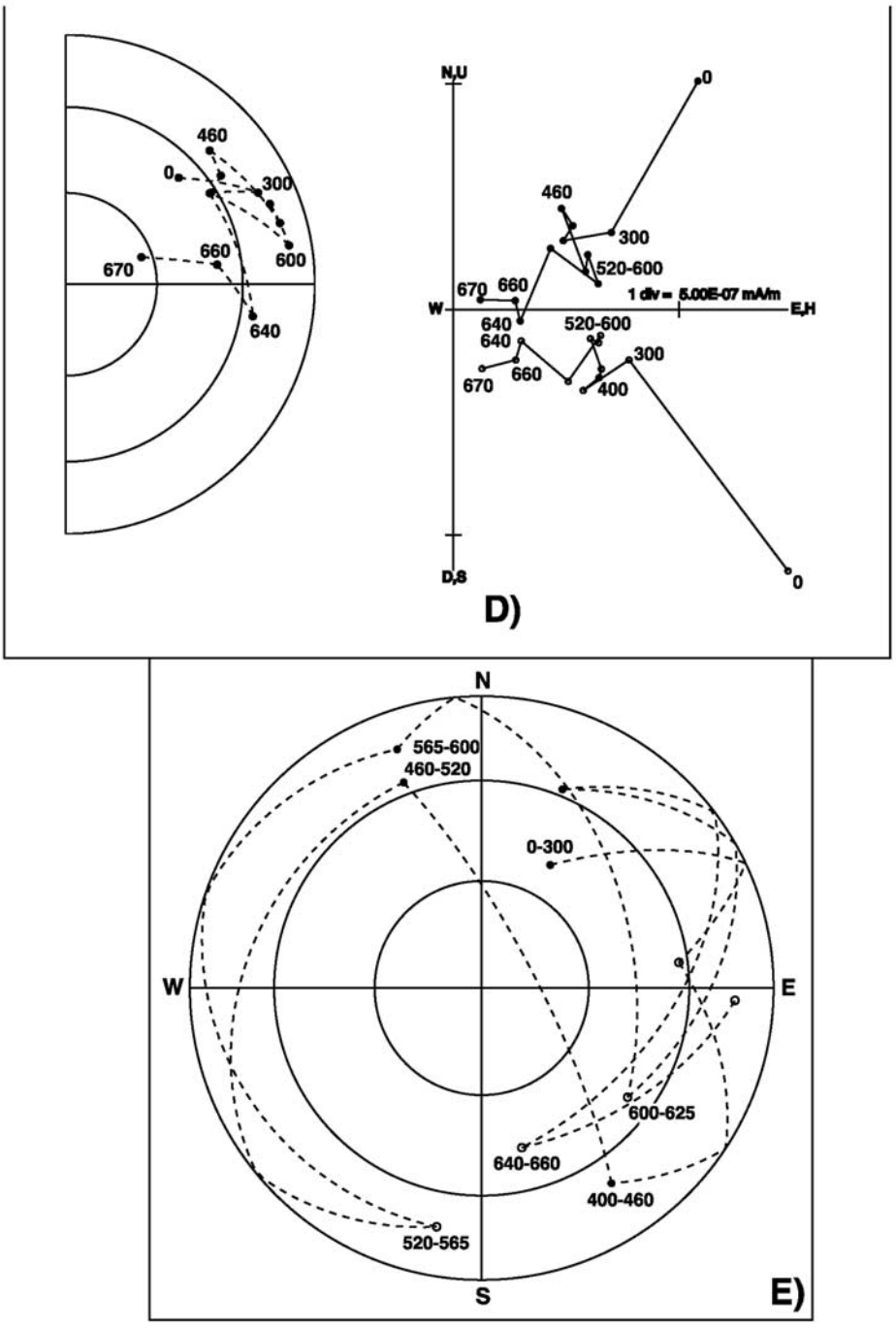


Figure 3. (continued)

they are located only 50 km apart. However, the thinner deposits at Romeroville are directly underlain by limestones of the Todilto Formation, whereas the Todilto Formation is absent under the thicker Summerville sequence at Trujillo hill, suggesting, in agreement with the general Summerville depositional pattern, that the farther to the east, the more marginal are localities to the extant Jurassic seaway of the time [e.g., *Imlay*, 1980]. Likewise, the thick sequence of channel sandstones overlying the Summerville red beds at Romeroville, in contrast to their thinness at Trujillo hill, again is consistent with the Trujillo locality having been located more marginally than Romeroville to the main depositional basin of the time.

[14] Despite these lithologic variations over such a short distance, the magnetostratigraphy of the Summerville sections indicates a definite correlation between the two sequences. The thin normal polarity intervals interspersed in predominantly reversed polarity at Trujillo are all similar in length and less than 1 m thick, except one. In the upper part of the Trujillo section, one normal polarity

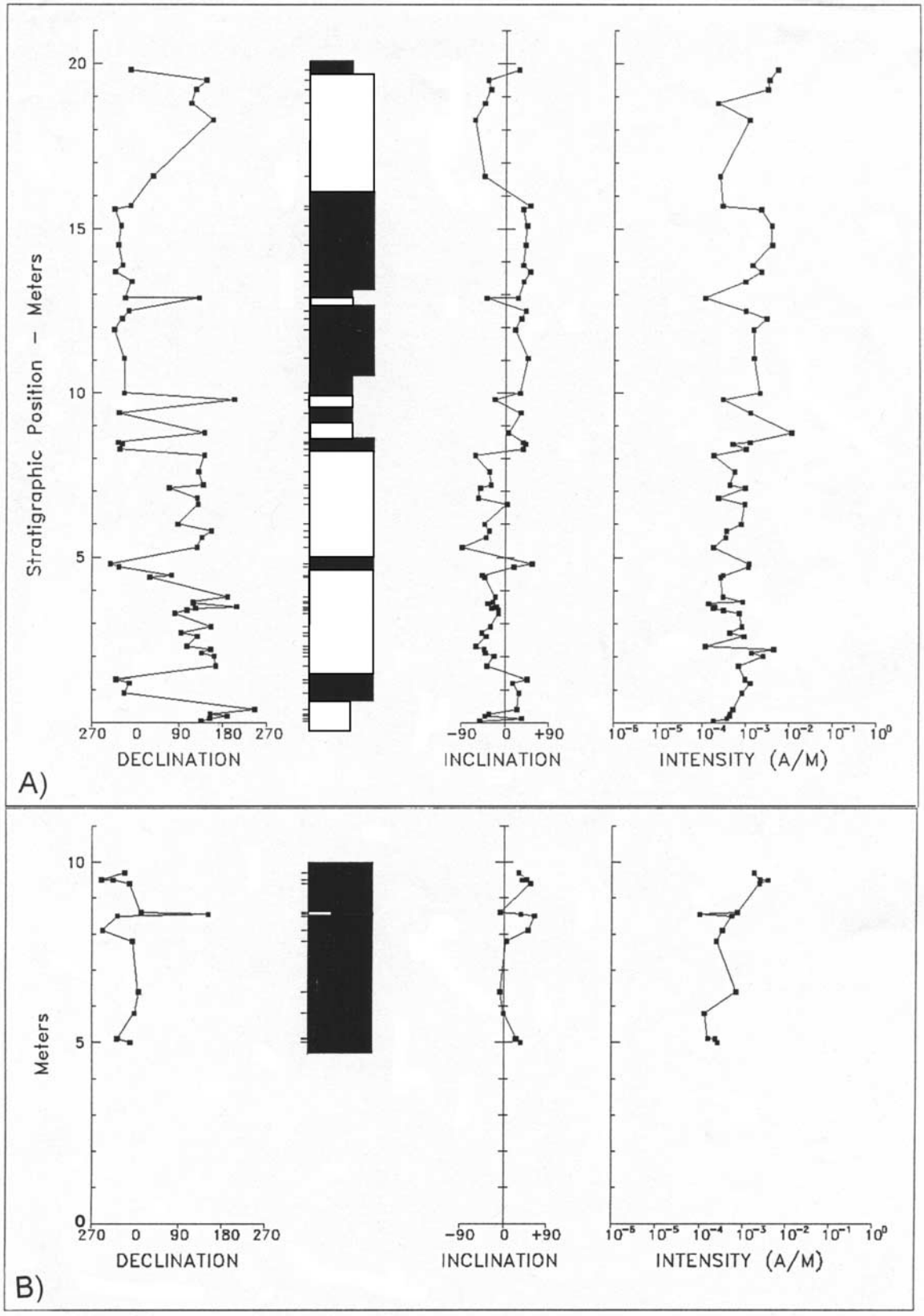


Figure 4. Summerville magnetostratigraphy: (a) Trujillo hill; (b) Romeroville. Black represents normal polarity, stippled represents reversed polarity, and polarity patterns extending only part way across the width of the polarity column designate less certain polarity regions.

Table 1. New Mexico Summerville Paleopole Analyses[a]

	Dec	Inc	*N*	*R*	*k*	α_{95}	Long E	Lat N	dP	dM	Colat
					Trujillo Mean Directions						
Group 1	326.2	36.8	20	19.37	30.3	6.0	147.4	56.8	4.1	7.0	69.5
Group II	329.8	30.1	18	17.05	17.8	8.4	137.3	56.9	5.2	9.3	73.8
Groups I & II	327.9	33.7	38	36.35	22.4	5.0	142.5	56.9	3.3	5.7	71.6
					Trujillo Reversal Tests						
Group 1 N	330.1	35.1	12	11.80	55.3	5.9	141.5	59.1	3.9	6.8	70.7
Group 1 R	139.7	−39.2	8	7.63	19.1	13.0	155.4	52.6	9.3	15.6	112.1
McFadden Classification = C											
Group II N	335.3	33.7	8	7.78	33.2	9.9	133.6	62.3	6.4	11.3	71.6
Group II R	145.5	−27.0	10	9.34	13.6	13.6	139.6	52.5	8.0	14.8	104.3
McFadden Classification = C											
Gps I&II N	332.2	34.5	20	19.57	44.1	5.0	138.5	60.4	3.3	5.7	71.0
Gps I&II R	143.1	−32.5	18	16.86	14.9	9.2	146.3	52.8	5.9	10.4	107.7
McFadden Classification = C											
					Romeroville						
Group I	330.1	41.2	6	5.72	17.6	16.4	148.2	61.6	12.2	20.0	66.3
Gps I&II	332.3	36.3	10	8.68	6.8	20.0	139.6	61.3	13.6	23.3	69.8
					VGP Pole Calculations						
Trujillo (Gp I)			20	19.39	31.0	6.0	148.2	57.3			
Romeroville (Gp I)			6	5.73	18.7	15.9	150.9	54.9			

[a] All directions are tilt corrected. *N* is the number of samples; *R* is the length of the resultant vector; *k* represents the precision of the sample populations [*Fisher*, 1953]; α_{95} is the half-angle limits of 95% confidence; Long E and Lat N are the east longitude and north latitude of the paleopole locations; dP and Dm and the semi-axes of the ellipse of confidence around the paleopole; Colat is the angular distance from the site to the paleopole; Gp(s) = Group(s). Reversals test used the analysis of *McFadden and McElhinny* [1990]; Classification C indicates that the normal and reversed means were between 10° and 20° apart, still a positive reversals test within the 95% confidence limits.

interval occupies 5.5 m of stratigraphic section. Similarly, the normal polarity sequence at Romeroville also is 5 m thick. It is highly likely that these two are correlatives, both because of the notable thinness of all other Trujillo normal polarity intervals, and also because the relative differences in the proximity of the two sites to the margin of the Mid-Jurassic seaway. That difference makes it likely that the Todilto limestones at the base of the Romeroville sequence are equivalent to the lowest Summerville red beds at the Trujillo locality. Located farther from the seaway, the marginal marine sabhka and terrestrial red bed deposition would have begun earlier at the Trujillo location while the seaway still occupied the Romeroville location.

[15] In the Trujillo section, six individual samples indicate changes of polarity not corroborated by any adjacent sample. In general these are Group I samples, making less likely the possibility that their remanence is a result of complete remagnetization. Very high reversal frequency during this time period is indicated by numerous low-amplitude, short-duration magnetic anomalies preserved in seafloor of this same age [*Handschumacher et al.*, 1988; *Sager et al.*, 1998]. The inferred high reversal frequency averages one reversal every 50,000–75,000 years [*Sager et al.*, 1998], suggesting that the single-polarity samples in the Summerville magnetostratigraphic record may record short-lived geomagnetic field polarity changes. Further support for a high reversal frequency during Summerville deposition is found in the magnetic behavior of the 34% of the sample population which exhibits removal of both normal and reversed polarity of apparently Jurassic directions (Group III); this characteristic of the remanence may indicate that either the magnetizing process was too slow to individually record the most rapid polarity fluctuations or that a single 2.5 cm paleomagnetic sample from this sedimentary setting is too large to document polarity changes of this frequency. A denser sampling and smaller samples are needed to test these speculations.

3.2. Summerville Paleopoles

[16] Tilt-corrected mean directions and corresponding pole positions were computed only from Groups I and II samples, using Fisherian statistics (Table 1). At Trujillo hill, separate paleopoles were computed from the Group I (demagnetization behavior univectorial to the origin of orthogonal axes plots, and MAD values of $\leq 4°$) and Group II (MAD values of 5°–10°, samples with smaller MAD values but linear behavior not directed to the origin, or samples displaying linear-to-the-origin behavior and low MAD values but exhibiting directions 2 standard deviations removed from the Group I mean). Samples with this latter Group II remanence characteristic were placed in the second group because such is the expectation of samples possessing depositional inclination error, orientation errors, or high-stability secondary remanences. The Group II pole is displaced 10° east of the Group I pole, although the large 95% confidence limits of each pole just include the position of the other pole.

[17] The degree to which the normal and reversed directions are antipodal was examined for both Groups I and II.

The mean directions are antipodal, but at the upper limits of statistical validity for the 95% confidence level [*McFadden and McElhinny*, 1990] (Table 1), designated by the reversal test as ''Class C'' (angle between mean directions is between 10°–20°). Neither group, nor a combination of Groups I and II, formed populations with more statistically significant antipodal means; this most likely reflects unremoved secondary magnetizations.

[18] Two mean directions were computed for the Romeroville locality, using criteria similar to that of the Trujillo calculations. The more poorly magnetized Romeroville strata produced only four samples with demagnetization trends which could be fit by lines with MAD values of $\leq 4°$ (Group I); to modestly improve statistics, two Group II samples with MAD values of $\leq 10°$ were included in the calculation of a Group I mean. Those six samples produced a mean direction statistically identical at 95% confidence to that from Trujillo hill (Figure 5a; Table 1), albeit with very large confidence limits as attend such a small sample population. The addition of the four other Group II samples (MAD values of >10°) to the Group I population substantially degraded the precision parameter (k), and also the α_{95} confidence limits (Groups I and II, Table 1).

[19] Despite the significant difference in structural attitudes of the two localities (nearly flat-lying strata versus strata dipping 35°), the Group I poles from both localities are statistically identical (Figure 5a) after structural correction. This agreement suggests that these samples reliably indicate the paleo-direction of the field which prevailed during Summerville deposition. The pole considered most representative of late Callovian North America is that calculated from only the Trujillo site, because combining the Romeroville data with the Trujillo data would inequitably emphasize one portion of the depositional sequence. Further, the Group I data provide the most representative pole calculation because their mean has half the dispersion (k, Table 1) of the Group II mean. Therefore, the best representation of the cratonic Summerville paleopole is considered to be that derived from Trujillo locality Group I data.

3.3. Callovian-Oxfordian Jurassic Polar Stillstand and the J-5 Unconformity

[20] For better comparison of this new Summerville paleopole to the directly overlying lower Morrison Formation paleopole, the New Mexico Morrison paleopoles [*Steiner et al.*, 1994] were recalculated from VGPs (Table 2). The Morrison Formation consists of two distinct facies which divide the formation into two members, and which yield different paleopoles, a fact observed both in Colorado in New Mexico [*Steiner et al.*, 1994; *Steiner*, 1998]. Recalculation of the lower Morrison paleopole employed all of the same data used in the 1994 analysis; recalculation of the upper Morrison pole excluded four samples that had been used in the earlier pole calculation, because even though univectorial-to-the-origin, their characteristic directions lay more than two standard deviations from the mean of the rest of the group. Such samples in the current Summerville analysis are considered Group II category rather than Group I, therefore these four were excluded in the recalculated Morrison pole. In the course of these recalculations, a separate pole was calculated for the greatly reduced thickness, but physically identical, lower Morrison Formation member at the Trujillo hill section (Table 2), a calculation not made in the earlier study because of the limited stratigraphic thickness and small number of samples [*Steiner et al.*, 1994]. The lower Morrison Trujillo paleopole is identical to the Romeroville lower Morrison pole (T and R in Figure 5b). As with the Summerville poles, the considerable difference in structural attitude (5° versus 35° dip) makes this result significant.

[21] The Summerville poles are statistically similar to the lower Morrison paleopoles (Figures 5c and 5d), both in Colorado-Utah and in New Mexico (95% and 90% confidence levels, respectively). The fact that the Summerville and lower Morrison Formations yield approximately the same paleopole indicates one of three possibilities: either (1) little time elapsed between deposition of the two formations, or (2) the North American (NA) plate (and at that time, the linked Colorado Plateau block) were stationary, or (3) these plates were rotating about an Euler pole essentially coincident with the geomagnetic pole. The fact that both formation poles occur at a bend in the NA apparent polar wander (APW) curve may favor possibility (2) over (3); a pause in plate motion is logical during the reorganization of plates that led to this change in the direction of the APW curve.

[22] The coincidence of Summerville and lower Morrison paleopoles was surprising because of the widely recognized stratigraphic break normally reported to exist between the two formations, the J-5 unconformity [*Pipiringos and O'Sullivan*, 1978; *Peterson*, 1988; *Anderson and Lucas*, 1992]. Nonetheless, the possibility that the J-5 unconformity represents minimal elapsed time is supported by published descriptions of the contact between the Summerville and Morrison Formations. In summarizing these descriptions, *Imlay* [1980] noted that descriptions through time alternately characterized the contact as either unconformable or gradational. He concluded that if all of these descriptions were correct, deposition of the Summerville Formation in some places was succeeded immediately by the Morrison Formation deposition, whereas in nearby locations minor erosion occurred prior to Morrison deposition [*Imlay*, 1980, p. 96]. Such a scenario is entirely compatible with the statistical similarity of the paleopoles above and below the unconformity. Essentially the same paleopole in both the Summerville and Lower Morrison Formations leads to a tentative conclusion that little or no plate motion of North America occurred between the late Callovian and Oxfordian, a period of possibly 6 m.y.

3.4. Rotation of the Colorado Plateau

3.4.1. Summerville Poles

[23] Despite the fact that both the New Mexico and the Colorado-Utah sampling sites indicate that the same paleopole prevailed during Summerville and lower Morrison

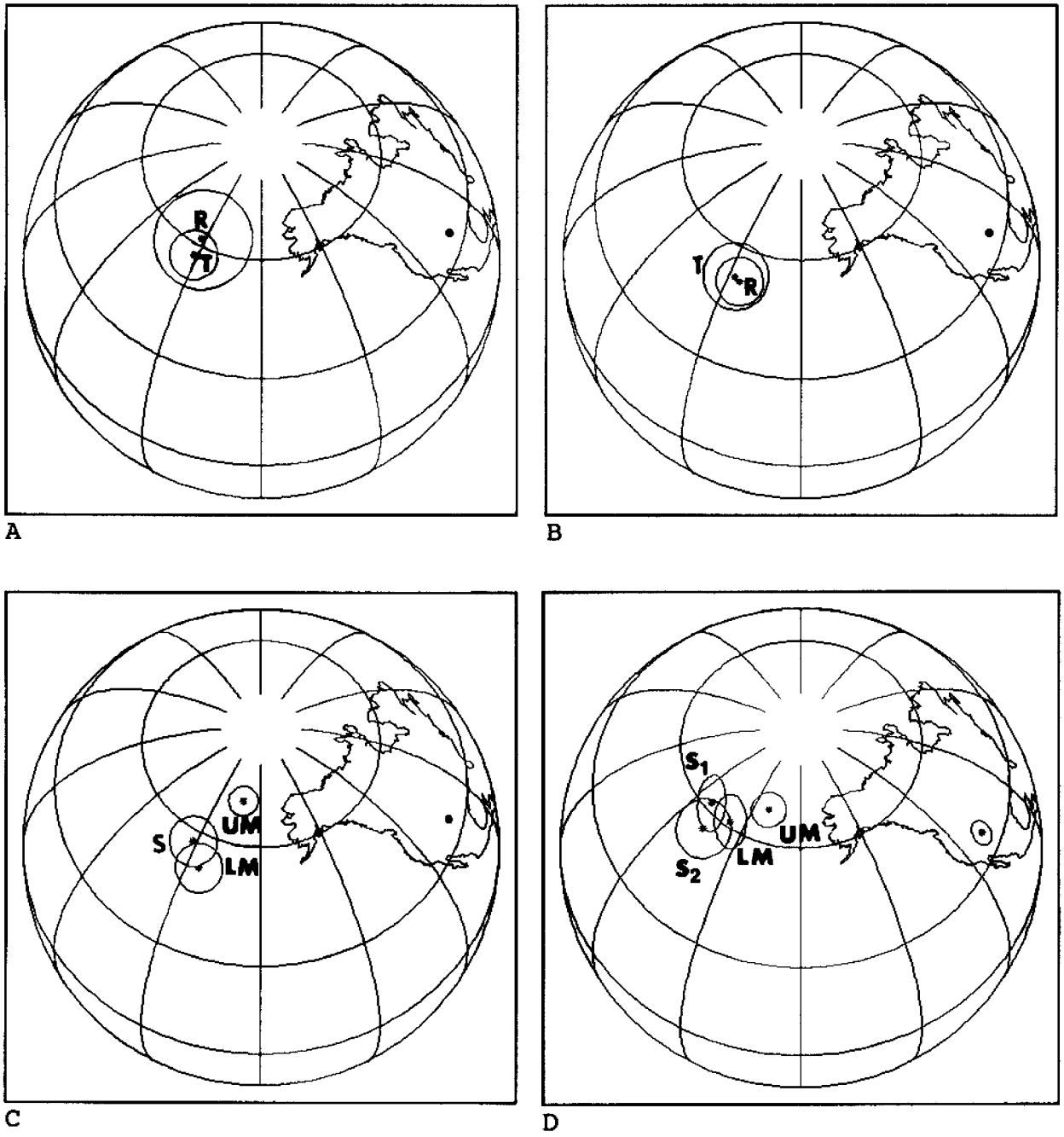


Figure 5. (a) Summerville paleopoles from the Trujillo (T) and Romeroville (R) localities. (b) Lower Morrison (Salt Wash) paleopoles from the Trujillo (T) and Romeroville (R) localities. (c) Cratonic Summerville (S) and lower (LM) and upper (UM) Morrison paleopoles. (d) Colorado Plateau Summerville poles (S1 = *Steiner* [1978]; S2 = *Bazard and Butler* [1992]) and lower (LM) and upper (UM) Morrison paleopoles.

time, the different sites give different actual positions for the paleomagnetic pole of that time. The New Mexico sites are located east of the Rio Grande Rift, presumably part of cratonic North America, whereas the Colorado-Utah sites reside on the Colorado Plateau which has been shown to have moved independently of the craton [*Steiner*, 1984; *Irving and Strong*, 1985; *Steiner*, 1986; *Bryan and Gordon*, 1986; *Steiner*, 1988; *Bryan and Gordon*, 1990; *Kent and Witte*, 1993; *Molina-Garza et al.*, 1998]. The Colorado Plateau Summerville and lower Morrison poles are displaced clockwise from the cratonic poles, and in an amount similar to other paleopoles derived from coeval strata

Table 2. Coeval Cratonic North American and Colorado Plateau Paleopoles

	N[a]	R	k	95% Conf.	Long E	Lat N	Reference
Late Pennsylvanian							
Craton L Penn Mean	*4	3.996	862.3	3.1	130.8	43.2	(2, 3, 4)[b]
Plateau Supai	45	43.94	41.6	1.7, 3.4	119.2	47.3	2
Early Permian							
Craton Mean of 3 [Cas., Ingle., Abo (see text)]	3*	2.996	455.4	5.8	121.8	49.2	(2,4,5)[b]
Pictou pole	1*	–	–	1.8, 3.6	125.5	42.1	6
Craton E Perm Mean of 4	*4	3.989	274.0	5.6	122.8	47.5	(2, 4, 5, 6)[b]
Plateau Cutler	66	–	–	2.3	110.7	50.5	7
Early Triassic							
Craton E Tr Mean	*8	7.992	927.8	1.8	114.9	46.1	(8,9)[b]
Plateau Moenkopi Mean	*4	3.996	689.8	3.5	101.2	56.7	(8)[b]
Late Triassic							
Craton Chinle Gp Mean	*5	4.992	531.2	3.3	81.7	57.5	10
Plateau Chinle Gp Mean	*4	3.999	2635	1.8	66.6	57.2	10
L. Mid-Jurassic							
Craton Summerville	20	19.39	31.0	6.0	148.2	57.3	1
Plateau Summerville 1	33	31.70	24.6	3.6, 6.9	128.3	61.8	11
Plateau Summerville 2	*11	10.74	38.9	7.2	133.4	56.0	12
E. Late Jurassic							
Craton, Romeroville, Lower Morrison	57	52.06	11.3	5.9	154.9	52.0	(13)[b]
Craton, Trujillo, Lower Morrison	7	6.89	54.6	8.2	152.6	52.8	(13)[b]
Plateau Lower Morrison	*3(32)	3.00	549.4	4.0, 6.5	142.2	61.4	14
L. Late Jurassic							
Craton Upper Morrison	51	49.45	32.2	3.6	167.1	70.6	(13)[b]
Plateau Upper Morrison Mean	*2	1.999	3881	4.4	159.0	67.9	(14,15)[b]

[a] Starred values of N indicate numbers of individual paleopoles or sites used in the analysis; unstarred values indicate numbers of samples or stratigraphic horizons.

[b] References calculated in this study (parentheses give reference to original data); 2, *Steiner* [1988]; 3, *Symons et al.* [1989]; 4, *Diehl and Shive* [1979]; 5, *Diehl and Shive* [1981]; 6, *Symons* [1990]; 7, *Gose and Helsley* [1972] (recalculated by *Scott* [1975]); 8, poles listed in Table 1 of *Steiner* [1998]; 9, *Molina-Garza et al.* [1996]; 10, *Steiner and Lucas* [2000]; 11, *Steiner* [1978]; 12, *Bazard and Butler* [1992]; 13, calculated from the *Steiner et al.* [1994] directions recalculated as VGPs; 14, *Steiner and Helsley* [1974]; 15, *Bazard and Butler* [1994].

deposited both on the Colorado Plateau and the North American craton [*Steiner*, 1986; *Kent and Witte*, 1993; *Steiner et al.*, 1994; *Molina-Garza et al.*, 1998], thus providing yet another data set supporting the clockwise rotation of the Colorado Plateau.

3.4.2. Previous Analyses of Colorado Plateau Rotation

[24] Since it first was pointed out that paleomagnetic data indicated that the Colorado Plateau had been rotated relative to cratonic North America [*Steiner*, 1984], the amount of rotation has been a matter of dispute [e.g., *Irving and Strong*, 1985; *Steiner*, 1986, 1988; *Bryan and Gordon*, 1986, 1990; *May and Butler*, 1986; *Kent and Witte*, 1993; *Bazard and Butler*, 1994; *Molina-Garza et al.*, 1991, 1993, 1995, 1996, 1998]. Rotations of 10° or more were advocated by *Steiner* [1984, 1986], *Irving and Strong* [1985], and *Kent and Witte* [1993], whereas for some time much smaller rotations ($\geq$ 5°) were advocated by *Bryan and Gordon* [1986, 1990], *May and Butler* [1986], *Molina-Garza et al.* [1991, 1995, 1996], and *Bazard and Butler* [1994]. Disagreement over the amount of Plateau rotation is essentially due to the large amount of APW experienced by the North American plate during the time period for which both craton and Plateau paleopoles exist for comparison (Carboniferous through Cretaceous), and the methods of making those comparisons. Only precisely dated strata can be used for an accurate assessment of the rotation amount; otherwise even small differences in age will distort the amount of apparent Plateau rotation. One approach has been to compare only quite well dated formation paleopoles of the same ages [*Steiner*, 1984, 1986; *Steiner and Lucas*, 2000]. The statistical difficulties with this approach are obvious because few data pairs exist. However, previous approaches which obtained better statistics [e.g., *Bryan and Gordon*, 1986, 1990] yielded Plateau rotation magnitudes no larger than 5°, and never values comparable to those from coeval-formation paleopole comparisons.

[25] Recently a new, intermediate, rotation value has entered the debate; 7.4 ± 3.8° has been advocated [*Molina-Garza et al.*, 1998]. This analysis uses mean poles for seven broad discrete time periods, essentially the same rotation analysis technique as that of the individual formation pole comparisons but on a coarser timescale and with added statistical analyses. Of note is the fact that that rotation analysis yielded a value which is statistically indistinguishable from the formation pole comparisons [*Steiner*, 1986, 1988; *Steiner and Lucas*, 2000]; 7.4 ± 3.8° [*Molina-Garza et al.*, 1998] is not statistically different from 10 ± 3° [*Steiner and Lucas*, 2000] or 11 ± 4° [*Steiner*, 1986]. So it would appear that analyses of Colorado Plateau rotation value are now in the same ballpark.

[26] Nevertheless, comparisons of individual formation poles of demonstrably coeval ages suggest that a rotation estimate of 7.4° may be a little low, perhaps still biased by

Table 3. Statistics for the Rotations of Colorado Plateau Paleopoles to Craton Paleopoles[a]

Period	Ma	Paleopoles Compared	Rotation Values
Late Pennsylvanian	**~295**	Supai Gp Pole vs Craton Mean Pole (Table 2)	significantly different for 6° to 12° of rotation = **9.0 ± 3.0°**
Early Permian	**~275**	Cutler Fm Pole vs Craton Mean Pole (Table 2)	significantly different for 3° to 14° of rotation = **8.5 ± 5.5°**
Early Triassic	**~245**	Plateau Mean Pole vs Craton Mean Pole (see text)	significantly different for 11° to 13° of rotation = **12.0 ± 1°**
Late Triassic	**~200**	Plateau Chinle Gp Mean vs Craton Chinle Gp Mean [*Steiner and Lucas*, 2000]	significantly different for 4.5° to 13° of rotation = **8.8 ± 4.2°**
late Middle Jurassic	**~161**	1. Plateau Summerville Fm (*Bazard and Butler* [1992]) vs Craton Summerville Fm	significantly different for 3° to 5° of rotation = **4.0 ± 1.0°**
		2. Plateau Summerville Fm (*Steiner* [1978]) vs Craton Summerville Fm	significantly different for 7° to 14° of rotation = **10.5 ± 3.5°**
early Late Jurassic	**~150**	Plateau Lower Morrison vs Craton Lower Morrison	significantly different for 7° to 18° of rotation = **12.5 ± 5.5°**
late Late Jurassic	**~148**	Plateau Upper Morrison vs Craton Upper Morrison	significantly different for +1° to −7° of rotation = **−3.0 ± 4.0°**

[a] All rotations quoted at the 95% confidence level except for "Early Triassic," quoted at 90% confidence.

the averaging of poles in large time blocks during times of rapid APW. However, the bias may come from the inclusion of a number of poles which should not have been included in the data set used by *Molina-Garza et al.* [1998]. Seven categories of poles should not have been used, about 21 paleopoles; because some length of description is required to explain the problems with the various poles, this topic is discussed in Appendix A.

[27] Following is an analysis of Colorado Plateau rotation which differs from the recent *Molina-Garza et al.* [1998] analysis in that only poles from formations deposited simultaneously on both the Colorado Plateau and the North American craton are used, and only from formations which can be shown to be coeval through magnetostratigraphy or biostratigraphy. Magnetostratigraphy now allows the establishment of coeval ages of strata at the 100,000–250,000 year level.

[28] A frequent criticism leveled at comparisons of Plateau-craton paleopoles from a single formation or at misfits between certain pole pairs when CP poles are rotated back [*Molina-Garza et al.*, 1998] has been that the ages of the respective formations probably are not exactly the same [e.g., *Bryan and Gordon*, 1990; *Molina-Garza et al.*, 1991, 1998]). The "age difference" concept is a convenient and overworked explanation; therefore, known details of the ages of all formations whose paleopoles are used in the rotation evaluation here are discussed in the following evaluation of CP rotation.

3.4.3. Analysis of Plateau Rotation Within Tightly Constrained Geologic Time Intervals

[29] Pairs of same-age Colorado Plateau and the North American craton paleopoles exist now for seven different time periods. Some are means of numerous studies and some are still individual poles (Table 2). The statistics developed by *McFadden and Lowes* [1981] and employed by *Kent and Witte* [1993] were applied to these seven pairs of poles to assess the amount of rotation of Colorado Plateau poles among coeval poles of narrowly constrained time intervals.

[30] The Euler pole used for the rotation evaluation was that of *Hamilton* [1988], located at 34.6°N, 254.6°E. Hamilton derived this pole by assessment of the structural data associated with the Laramide Orogeny, but Euler poles for the later opening of the Rio Grande Rift [*Hamilton*, 1981; *Cordell*, 1982] are quite similar. Moreover, the deformation associated with the Pennsylvanian-Permian orogeny produced structures within the Colorado Plateau crust that are so similar to Laramide structures that it is difficult to separate the former from the latter [*Hamilton*, 1978], implying a similar Euler pole for the Late Paleozoic deformation involving the Colorado Plateau. Finally, use of any of *Hamilton*'s [1978, 1981, 1988] Euler poles does not change any result by more than 0.3°. Moreover, rotations utilizing a wide range of Euler poles located within the general region of the Texas-eastern New Mexico-Colorado vicinity all bring Plateau and craton paleomagnetic poles into statistically reasonable coincidence at a 90–95% confidence level. *Molina-Garza et al.* [1998] used this same pole in their final analysis, because it is geologically reasonable and their statistical search for a best fit Euler pole yielded a, as far as is known at present, geologically meaningless rotation pole, and one with exceedingly large confidence limits.

[31] Using *Hamilton*'s [1988] Euler pole, the amount of rotation was assessed by paleopole comparisons individually in each age group; the method was a systematic incremental change of latitude and longitude (separately) by 1° increments, then a narrowed search at 0.1° increments. Fisherian statistics were used in the analysis. The resulting mean displacement value and the range of displacement values permitted within 90% to 95% probability limits for each of the seven paleopole groups are listed in Table 3. The confidence limits quoted in Table 3 are based on the amount of rotation needed to move both paleopoles of a pair into of

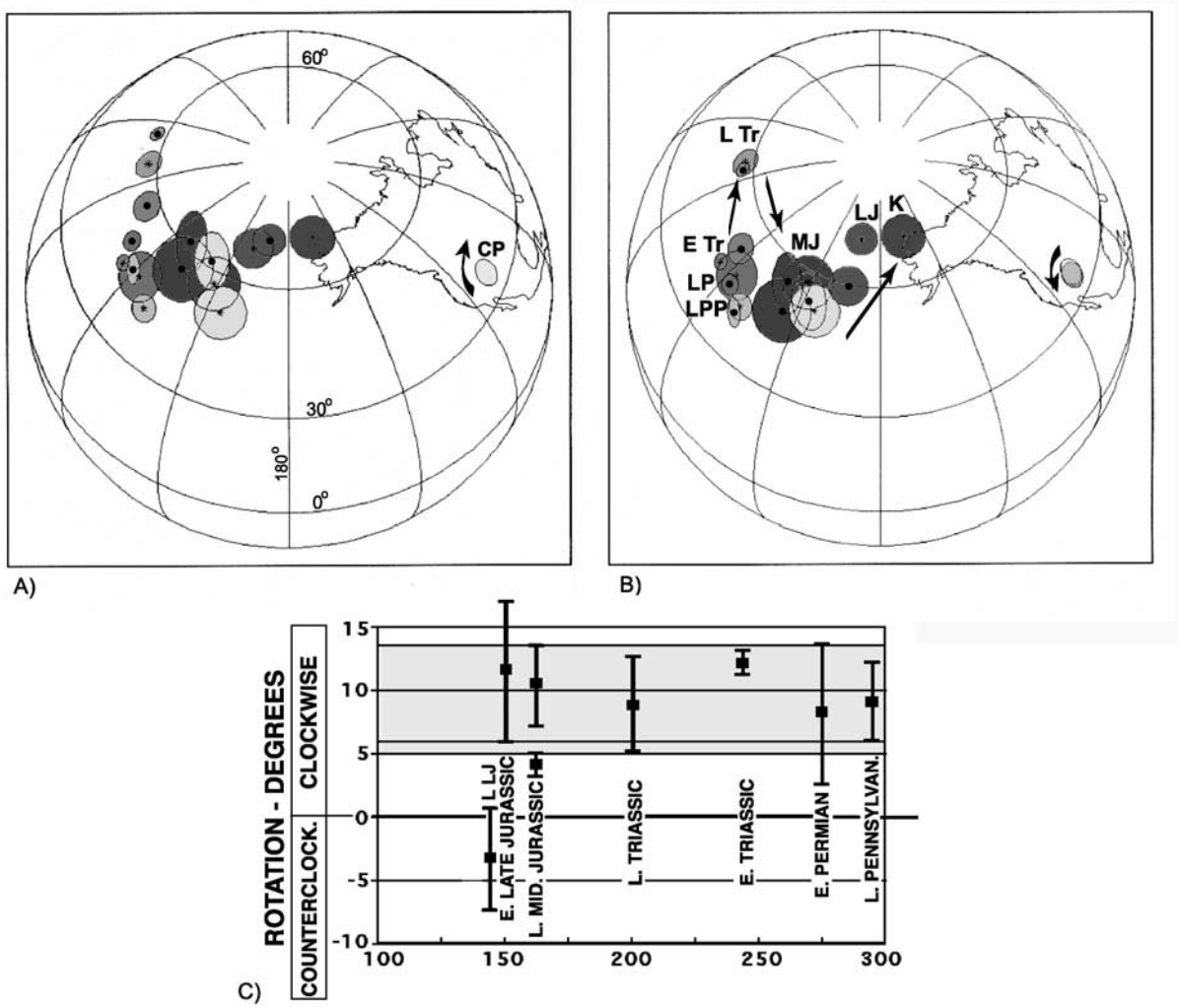


Figure 6. (a) Coeval paleopoles from the Colorado Plateau (solid dots) and stable NA craton (asterisks): Late Pennsylvanian (LPP, light blue); Early Permian (EP, rose pink); Early Triassic (ETR, orange); Late Triassic Chinle Group (LTR, green); Late Middle Jurassic, Callovian, Summerville (MJ, blue); early Late Jurassic, Oxfordian-lower Kimmeridgian lower Morrison Formation (yellow); and late, Late Jurassic, Kimmeridgian-Tithonian upper Morrison Formation (red). For reference, the averaged NA Late Cretaceous pole (Globerman and Irving, 1998) is shown in maroon. The Colorado Plateau is shaded gray. (b) Paleopoles of Figure 6a after the Colorado Plateau poles have been restored by a single 9° counterclockwise rotation around an Euler pole located at 34.6°N, 254.6°E. Arrows along the path APW path indicate the age progression of paleopoles. (c) The amount of rotation indicated by various Colorado Plateau paleopoles plotted against paleopole age; clockwise (counterclockwise) rotation is positive (negative). See color version of this figure at back of this issue.

each other's respective 95% confidence limits. Figure 6a shows the paleopoles pairs prior to rotation.

3.4.3.1. Late Carboniferous

[32] During the time interval of the Missourian through Virgilian Stages, no paleopole change is discernable among North American cratonic data. Lack of reversal of the geomagnetic field renders magnetostratigraphy useless in this time period, but biostratigraphic ages are determined from fusulinids, a particularly reliable biostratigraphic group for the Late Pennsylvanian and Permian. The Colorado Plateau is represented by a single pole from the Wescogame Formation (Supai Group) of latest Pennsylvanian, Virgilian age. The Wescogame age was determined from fusulinids and corals [*McKee*, 1982]. Four Late Pennsylvanian paleopole determinations have been made from cratonic NA: two are poles from the Casper Formation (Colorado and Wyoming), one of the just older Desmoinian Stage and the other of Missourian-Virgilian age; both ages are determined by fusulinid biostratigraphy

[*Thomas et al.*, 1953]. The third craton pole is from the Laborcita Formation (eastern New Mexico); its age is determined as post-Virgilian, pre-Early Permian age by fusulinid systematics [*Steiner and Williams*, 1968; *Ross and Ross*, 1998]. The fourth craton pole is that of cycle 1 of the Pictou Group (Newfoundland), also of Missourian to Virgilian age from palynology [*Symons et al.*, 1989]. Interestingly, three of these four craton poles are identical at the >99% probability level; the fourth lies within the 95% confidence limit of the other three. A mean of the four (Table 2) has high R and k values, supporting a paucity of APW during this time. At the 95% limit of confidence, the Plateau Wescogame paleopole is displaced 9° ± 3° clockwise from the cratonic Late Pennsylvanian mean pole (brown poles in Figure 6a).

3.4.3.2. Early Permian

[33] A Cutler Formation paleopole was calculated from Colorado Plateau strata [*Gose and Helsley*, 1972; modified by *Scott*, 1975]. The formation is Wolfcampian (and possibly earliest Leonardian) in age, based upon interfingering relationships with fossiliferous marine limestones [*Baars*, 1962, 1987]. The Cutler Formation is laterally traceable into New Mexico; in central New Mexico, the same arkosic units shed from the Uncompaghre Highland are called the Abo Formation [*Baars*, 1962, 1987]. The Abo Formation intertongues with the fusulinid-bearing, middle and late Wolfcampanian (and possible early Leonardian) Hueco Limestone [*Pray*, 1952; *Otte*, 1959; *Williams*, 1963]. Exposures east of the Rio Grande Rift yielded an Abo paleopole [*Peterson and Nairn*, 1971]; these were restudied [*Steiner*, 1988] and an essentially similar, but better demagnetized paleopole was obtained.

[34] Two cratonic (Wyoming and northern Colorado) Wolfcampian formations also have yielded paleopoles, the Ingleside Formation and the Wolfcampian portion of the Casper Formation [*Diehl and Shive*, 1979, 1981]. The ages of both are well determined from fusulinids, conodonts, and brachiopods from marine limestone beds within these formations (summarized by *Diehl and Shive* [1981]). A fourth cratonic Wolfcampian paleopole was derived from northeastern North America, from the upper Pictou beds (cycle 4) of Prince Edward Island [*Symons*, 1990].

[35] *Molina-Garza et al.* [1991, 1998] have expressed concern that the Abo paleopole may not representative of the Early Permian of stable North America (NA); in fact, that pole is not statistically distinguishable from other Early Permian paleopoles from the North American craton. *Molina-Garza et al.* [1991] noted that the Abo pole had a slightly more westerly longitude (4 ± 2°) than the upper Pictou and upper Casper poles and worried that it may have been subjected to some local rotation. However, no structural evidence exists [*Kelley*, 1979; *Chamberlin and Anderson*, 1989] that would be substantiate this speculation of local rotation. Moreover, statistical analysis indicates that the Abo paleopole agrees better with the Ingleside and upper Casper poles than does the upper Pictou pole, and that among the four Early Permian cratonic paleopoles, the Pictou pole is marginally the outlier in the group. The Pictou paleopole does not lie within the α_{95} circle of confidence attending a mean of the other three poles (Table 2), and its 95% confidence limits barely overlap those of that mean (Table 2).

[36] The difference between the Pictou pole and the other three Early Permian poles may be compatible with a counterclockwise rotation of the Pictou sampling region, a scenario suggested because of its similarity to that between the Late Triassic Newark Basin paleopoles and the coeval cratonic Chinle paleopoles. The paleopoles of the magneto- and biostratigraphically same-age Chinle and Newark Groups are separated by 9 ± 3°, a discrepancy suggesting a counterclockwise rotation of the Newark Basin with respect to the craton [*Steiner and Lucas*, 2000]. The same sense of discrepancy between the Pictou and the other cratonic North American poles may suggest that more of the eastern margin of North America has experienced rotation relative to the cratonic interior than we recognize at present, and that the Pictou pole may not be representative of the NA craton.

[37] But statistically, none of the four presumably cratonic Early Permian North American paleopoles lies further from each other than another; therefore without additional evidence, they must be considered a statistical group, and their mean provides the cratonic Early Permian paleopole (Table 2). The validity of this pole as representative of cratonic North America has recently been confirmed by a Permian paleoclimate and paleolatitude study; five new paleopoles of progressively younger Early Permian (Wolfcampian) age strata were produced from sites in central Texas (M. Steiner, ms in preparation). These Texas paleopoles are quite tightly grouped and agree exceedingly well with the craton poles and best with the Abo and upper Casper paleopoles. In fact this agreement removes the *Molina-Garza et al.* [1998] concern that the Abo pole does not represent cratonic North America.

[38] Rotations between 3° and 14° (Table 3) bring the Plateau mean paleopole into the 95% confidence limits of the mean craton paleopole. The Early Permian analysis thus indicates that the Colorado Plateau paleopole has been rotated 8.5 ± 5.5° relative to that of the craton (purple poles in Figure 6a).

3.4.3.3. Early Triassic

[39] Early and Middle Triassic is the next younger time period for which coeval strata exist on both the craton and Colorado Plateau. The Early Triassic Moenkopi Formation unconformably overlies the Early Permian Cutler strata of the Colorado Plateau. On the craton of northeastern Colorado and Wyoming, the Red Peak Formation (Chugwater Group) was deposited. In cratonic eastern New Mexico, the younger part of the Moenkopi Formation also was deposited. The dominantly terrestrial Moenkopi strata intertongue in the western Colorado Plateau region with marine limestones containing Smithian and Spathian ammonites [*Shoemaker and Newman*, 1959]. Above the ammonite-bearing limestones, the strata become entirely terrestrial, but contain a diagnostic vertebrate amphibian which is also found in marine deposits in Europe and

which dictates an early Anisian (earliest Middle Triassic) age [*Lucas and Morales*, 1985; *Morales*, 1987; *Lucas and Hayden*, 1989]; both Plateau and craton deposits contain this amphibian.

[40] Numerous magnetostratigraphic studies of the Moenkopi Formation have defined its magnetostratigraphy exceedingly well [e.g., *Helsley and Steiner*, 1974; *Purucker et al.*, 1980; *Molina-Garza et al.*, 1991; *Steiner and Lucas*, 1992; *Steiner et al.*, 1993]. Comparison of Moenkopi magnetostratigraphy with global marine magnetostratigraphy indicates that the Moenkopi spans approximately the older stage designation "Dienerian" through the Anisian Stages of the Early and Middle Triassic [*Steiner et al.*, 1989, 1993]. Magnetostratigraphy also shows that the cratonic Red Peak Formation [*Picard*, 1964; *Herrero-Brevera and Helsley*, 1983; *Shive et al.*, 1982] is the same age as the lower and middle of the Moenkopi Formation of the Colorado Plateau [*Steiner et al.*, 1993]. Magnetostratigraphically also, cratonic eastern New Mexico Moenkopi strata [*Molina-Garza et al.*, 1991; *Steiner and Lucas*, 1992; *Steiner et al.*, 1993] are the same age as the upper part of the Colorado Plateau Moenkopi, a fact also demonstrated by vertebrate biostratigraphy of the two locations [*Lucas and Morales*, 1985; *Morales*, 1987; *Lucas and Hayden*, 1989].

[41] Biostratigraphically, the basal Red Peak Formation is constrained to be post-earliest Triassic by its interfingering relationship with the underlying marine Dinwoody Formation, which contains earliest Early Triassic (Griesbachian) conodonts [*Paull and Paull*, 1983]. The Red Peak strata are conformably overlain by the sparsely fossiliferous Alcova Limestone (of probable Spathian, or perhaps Smithian, age [*Picard*, 1997; *Lucas*, 1994]), over which the Jelm/Crow Mountain Formation complex of probable early Anisian age [*Lucas*, 1994] lies conformably. Thus, biostratigraphically, the Red Peak Formation is nominally "Dienerian," and perhaps Smithian.

[42] The Moenkopi and Red Peak Formation magnetostratigraphies are biostratigraphically dated by comparison to those of fossiliferous marine Early Triassic strata of China and the Canadian Arctic [*Steiner et al.*, 1993]. Nevertheless, the exactness of the age match is a moot issue, because all Moenkopi Formation studies have indicated a lack of apparent polar wander during Early and early Middle Triassic time. A meter by meter study of paleopoles through approximately 300 m of the Moenkopi Formation, encompassing 10 polarity intervals, showed no change of paleopole [*Helsley and Steiner*, 1974]. The early Middle Anisian Plateau Moenkopi poles are identical to those of the Dinerian-Smithian-Spathian portions of the formation [*Steiner*, 1998] (Table 2). On the craton, six identical Early Triassic paleopoles derived from the Red Peak Formation (Table 2 of *Steiner* [1998]) are statistically indistinguishable from two early Middle Triassic poles from the Anton Chico Member of the Moenkopi; a mean craton Early to early Middle paleopole was calculated for comparison to the Plateau pole (Table 2). Of importance is the fact that all Early Triassic studies confirm that North America did not change position relative to the paleopole throughout the Early and early Middle Triassic (minimally, post-Griesbachian through early Anisian time). (Hence, the contention of *Molina-Garza et al.* [1998] that age differences explain some of the angular separation of craton and Plateau poles is incorrect.)

[43] The mean paleopoles for the Colorado Plateau and craton are shown in orange in Figure 6a. The Colorado Plateau mean pole is displaced by 12 ± 1° with respect to the mean craton pole. The nearly exact age equivalence of the Plateau and craton formations and the lack of APW imply that the difference in paleopole position is due to appreciable rotation of the Colorado Plateau.

3.4.3.4. Late Triassic

[44] The Chinle Group overlies the Moenkopi Formation both on the Colorado Plateau and in cratonic New Mexico. It consists of numerous lithologies, hence a plethora of formations names. Despite its varied nature, a combination of litho-, bio- and magnetostratigraphy has established good correlation within the Group [*Lucas*, 1993; *Steiner and Lucas*, 2000]. Four biostratigraphic faunachrons were discriminated within the Chinle Group [*Lucas and Hunt*, 1993]. Paleopoles have been obtained from craton exposures of all four faunachrons and from three faunachrons of Colorado Plateau strata. Chron by chron comparisons of paleopoles were performed by *Steiner and Lucas* [2000], and are reiterated here to emphasize that poles from each faunachron individually favor a "large" rotation of the Colorado Plateau. At the 95% confidence level, the late Carnian (Adamanian faunachron) Plateau paleopole (Blue Mesa Member of the Petrified Forest Formation) [*Steiner and Lucas*, 2000] has been rotated 7.5 ± 1° clockwise of the cratonic Adamanian age paleopoles of the Garita Creek and Tecovas Formations [*Molina-Garza et al.*, 1995, 1996]. Paleopoles from the superjacent early Norian (Revueltian faunachron) of the Plateau, the Painted Desert Member of the Petrified Forest Formation [*Steiner and Lucas*, 2000] and the Owl Rock Formation [*Bazard and Butler*, 1991] are rotated 10.5 ± 4.5° from the cratonic Revueltian age Bull Canyon Formation paleopole [*Bazard and Butler*, 1991]. For late Norian–Rhaetian time (Apachean faunachron), two identical paleopoles from the Plateau Church Rock Formation [*Reeve*, 1975 (recalculated by *Gordon et al.* [1984]); *Kent and Witte*, 1993] are rotated clockwise 8.5 ± 4.5° from two cratonic paleopoles from the Redonda Formation [*Reeve and Helsley*, 1972 (recalculated by *Gordon et al.* [1984]); *Molina-Garza et al.*, 1996].

[45] Once again, however, the precision of the age matches is not an issue, because minimal apparent polar wander occurred during Late Triassic time; as demonstrated and figured by *Steiner and Lucas* [2000], three separate groups of paleopoles, those of the Newark Supergroup [*Kent and Witte*, 1993], the cratonic Chinle poles, and the Plateau Chinle paleopoles, all indicate little or no apparent polar motion. A mean Chinle paleopole for the Plateau is rotated 9 ± 4° clockwise from the mean craton Chinle pole [*Steiner and Lucas*, 2000] (green poles in Figure 6a; comparison was not made to the Newark poles because

those appear to be rotated with respect to cratonic Chinle poles) [*Steiner and Lucas*, 2000].

3.4.3.5. Late Middle Jurassic

[46] No coeval craton and Colorado Plateau paleopoles exist for the Early, nor most of Middle Jurassic time. The next younger contemporaneous paleopoles are those of the late Callovian Summerville Formation. The present investigation has produced cratonic paleopoles, and three previous studies have investigated Plateau Summerville strata.

[47] The Summerville remanence is puzzling for several reasons. In the 1970s, two studies of the formation on the Colorado Plateau were made. In the earliest study, only a short stratigraphic section (10 m) was sampled on the eastern flank of an anticlinal structure known as the San Rafael Swell [*Steiner and Helsley*, 1972]. The later study sampled essentially the entire formation, 120 m, on the western flank of the same structure, sampling from a recently made exposure, a new I-70 road cut [*Steiner*, 1978]. Both collections exhibited high dispersion among characteristic sample directions. The dispersion is far in excess of that normally encountered in paleomagnetic studies of terrestrial red sedimentary strata. The third study sampled the same locality as the 1978 study [*Bazard and Butler*, 1992]. High dispersion again was observed to be a characteristic of Summerville remanence.

[48] In the 1972 and 1978 studies, the samples which displayed the most linear behavior exhibited very similar characteristic directions after correction for the opposite dips of the strata at the two localities. Separately or combined, the two localities gave a paleopole near 110° east longitude and 67° north latitude (N = 15; R = 14.80; k = 71.07; α_{95} = 4.6), distinctly removed from poles resulting from inclusion of lesser quality data. Inclusion of lesser quality data (and doubling the sample population to N = 29) moved the pole to about 128°E, 62°N (R = 27.76; k = 22.52; α_{95} = 5.8) [*Steiner*, 1978]. Because of the peculiar difference between the most linearly behaved samples and those with less linear demagnetization trajectories, a total of four poles was calculated, specified by the widening angular distance encompassed by the sample directions, <25°, <37°, <90°, and <120° (Table 1 of *Steiner* [1978]). The two poles just quoted are the <25° and <37° poles; the two additional calculations only added an additional 10 samples, and did not appreciably change the mean or statistics. Subsequently, *Bazard and Butler*'s [1992] restudy of the same locality yielded a paleopole (133.4°E, 56.3°N) similar to the second of *Steiner*'s [1978] poles. *Bazard and Butler* [1992] also calculated multiple paleopoles (3) because of the dispersion within the data. Their three poles do not differ appreciably, but the dispersion was reduced (from k = 6 to k = 39) and 95% confidence limits, halved (15.6° to 7.4°). All of *Steiner*'s [1978] poles except that from the most restricted data set (lie within the 95% confidence limit of Bazard and Butler's most restricted data set pole.

[49] Now, in affirmation of the high-dispersion characteristic of Plateau Summerville data, the craton Summerville data of this study also exhibit high dispersion. Among the fourteen CP-craton paleopoles under consideration in the present investigation, the Plateau Summerville pole of *Bazard and Butler* [1992] has the highest dispersion, and the craton Summerville pole of this study has the second highest. These facts indicate that there is something peculiar about the remanent magnetization of the Summerville Formation.

[50] Exactly what the problem is with the Summerville remanence is remains to be investigated. It is highly likely that the large dispersion in remanence directions is related to the high frequency of geomagnetic reversals during this time. Both the Summerville magnetostratigraphic records and the marine magnetic anomaly records suggest a high reversal frequency. The high dispersion is likely attributable to a failure of the red bed magnetization process to keep up with the reversal rate. One likely scenario in terrestrial red beds is a first acquisition of detrital remanence in one polarity, followed quickly (within ~100,000 years) by a secondary chemical remanence in the next polarity interval. In such a scenario, the samples possessing the most linear demagnetization behavior are likely to be ones from intervals in which the field remained a bit longer than average in one polarity. If this scenario is the reason for the peculiar Summerville behavior, only samples possessing the most linear demagnetization trends to the origin of orthogonal axes plots will give the true geomagnetic field direction during Summerville time.

[51] The most recent study employed the technique of least squares fitting of lines to directions obtained from multiple demagnetization steps, and should have provided the better estimate of the Summerville paleo-direction. However, the unusually high dispersion experienced by all studies and the disparity of mean directions between *Steiner*'s [1978] most linearly behaved samples and the majority of samples remain to be explained. The poles of the 1978 study represent twice the stratigraphic section as do those of the 1992 study, although the value of this is unclear in the face of the dispersion abnormality of this remanence. Because of the uncertainties surrounding Summerville remanence, paleopoles from both Summerville studies are retained as potentially representative of the Plateau paleopole during Summerville time.

[52] Rotational evaluation of *Bazard and Butler*'s [1992] Plateau Summerville paleopole shows that it lies 10.0° counterclockwise from the New Mexico cratonic Summerville pole, although the large 95% confidence limits on both poles substantially reduce the statistical significance of this difference. Only 4 ± 1° of clockwise rotation brings both the Plateau and craton poles into the 95% confidence limits of the other (Table 3; blue poles of Figure 6a). Moreover, the direction of displacement is approximately perpendicular to that exhibited by other Colorado Plateau-craton pole pairs. In addition to the peculiarities of Summerville remanence just discussed, *Steiner*'s [1978] pole was retained because it does exhibit a considerably more similar direction of displacement from the craton Summerville pole to that of other Plateau-craton paleopole pairs. This may be entirely fortuitous, or it may have some meaning. The 1978 Plateau Summerville pole (the <37° pole) indicates 10.5° ± 3.5° of

rotation relative to the craton Summerville pole (Table 3; Figure 6a).

3.4.3.6. Late Jurassic

[53] Overlying the Summerville Formation, the Morrison Formation has yielded two pairs of paleopoles, both from the craton and the Colorado Plateau [*Steiner and Helsley*, 1975; *Steiner et al.*, 1994]. In both areas, the pronounced lithologic change which divides the formation (sandstone-dominated lower member versus variegated bentonitic mudstone upper member) is associated with statistically distinct paleopoles. The dual paleopoles have been observed from stratigraphic sequences separated by 425 km. The arc-distance between the upper and lower member paleopoles is similar on both the Plateau and craton: lower and upper Morrison poles of Colorado are 11 ± 5.3° apart [*Steiner and Helsley*, 1975], and New Mexico poles are 16 ± 4.5° apart [*Steiner et al.*, 1994]. (Contrary to the statement of *Molina-Garza et al.* [1998], I do not consider the cratonic lower Morrison pole to be a pole of low reliability.) Despite the similarity of dual paleopoles, the craton and Plateau do not give the same lower Morrison paleopole (yellow poles in Figure 6a). At the 95% confidence level, the lower Morrison poles indicate rotation of the Colorado Plateau pole by 12.5° ± 5.5°.

[54] The upper Morrison member has been investigated twice on the Colorado Plateau [*Steiner and Helsley*, 1975; *Bazard and Butler*, 1994] with statistically identical results; a mean pole was calculated from the two (Table 2). In contrast to the displacement between lower Morrison paleopoles, the upper Morrison poles are statistically indistinguishable between the craton and the Plateau (red poles in Figure 6a), indicating no clockwise rotation. At a 95% confidence limit, the upper Morrison poles permit 1° of clockwise rotation of the Colorado Plateau pole and up to 7° of counterclockwise rotation (a rotation of −3.0° ± 4.0°).

3.4.4. Critical Assessment of the Amount of Colorado Plateau Rotation

[55] The previous discussion demonstrated that six of the seven Mesozoic-Late Paleozoic paleomagnetic poles from the Colorado Plateau (CP) are displaced clockwise from coeval poles of cratonic North America (Figure 6a). Table 3 tabulates the rotations required to bring each of the seven CP paleopoles into statistical condidence with its cratonic counterpart; 9° to 12° of counterclockwise rotation around *Hamilton*'s [1988] Laramide Euler pole is needed to bring five (or six: Summerville uncertainty) of the paleopole pairs into coincidence. A single rotation of CP poles by 9° ± 3° brings these CP poles into the best statistical coincidence with coeval craton poles (Figures 6b and 6c).

[56] For three of the seven paleopole pairs, the 9° ± 3° rotation does not generate statistical coincidence at the 95% probability level. The problems surrounding one, the Summerville paleopoles, have already been discussed in detail. A second pair is the Early Triassic paleopoles; although the 9° ± 3° rotation brings them into longitudinal agreement, no rotations around this Euler pole can generate statistical coincidence of the CP and craton poles at any reasonable confidence level. After 9° of counterclockwise rotation, these poles still are statistically distinct at >80% confidence level. This result contrasts sharply with the fact that of the seven time periods, the Early Triassic formations have the best preserved (least overprinted) magnetization, some of the best determined magnetizations (number of studies on each tectonic block), and indisputable bio- and magnetostratigraphic age matches. The failure of the 9° counterclockwise rotation around the *Hamilton* [1988] Euler pole, to restore these poles suggests that an additional tectonic event has involved one of the tectonic blocks that yielded these Early Triassic poles. The preponderance of cratonic Early Triassic paleopoles was obtained from the Wyoming crust, thus raising the possibility that an unidentified rotation(s) accompanied the Laramide deformation of that crust. As discussed earlier, the Early and early Middle Triassic paleopoles are identical, and the early Middle Triassic portion of the Moenkopi Formation has been investigated in two localities in New Mexico; these yield paleopoles that are statistically identical to the Wyoming Early Triassic poles. Therefore, this enigma involving the Early Triassic paleopoles remains unresolved at this time. The third paleopole pair not responding favorably to a 9° counterclockwise restoration is that of the Late Jurassic upper Morrison poles; this discrepancy will be addressed shortly.

[57] Despite the fact that a single rotation brings many of the CP-craton paleopole pairs into coincidence, geologic data clearly demonstrate that clockwise rotation of the Colorado Plateau was not the consequence of a single event, but of at least three or more events. *Hamilton* [1978, 1981, 1988] has presented compelling geologic arguments for a Late Cretaceous/Early Tertiary clockwise rotation of the Colorado Plateau in association with the Laramide Orogeny. Subsequent Late Tertiary activity opened the Rio Grande Rift, causing a further small clockwise rotation of the Plateau [*Hamilton*, 1981; *Cordell*, 1982]. Furthermore, *Hamilton* [1978] also observed data indicating a Late Pennsylvanian rotation, although it could not be well defined because of the displacements of Late Pennsylvanian deformation were essentially the same as those of the Laramide Orogeny.

[58] *Hamilton*'s [1978, 1981] initial estimates were 4° of clockwise rotation of the Colorado Plateau as a consequence of Laramide deformation, and an additional 3° of rotation associated with opening of the Rio Grande Rift. But the cumulative 7° of clockwise rotation around Hamilton's Euler poles for the respective deformations did not restore the Plateau paleopoles to as tight a statistical coincidence with their cratonic counterparts as was possible. (Hamilton's Euler poles for the two events are quite close together, thus, are essentially the same.) A greater amount of total rotation seemed necessary [*Steiner*, 1986], and subsequently, *Hamilton* [1988] advocated the possibility of up to a total of 11° of Plateau rotation, based largely on the paleomagnetic evidence marshaled by *Steiner* [1986]. Rotation of the Colorado Plateau in response to the opening the Rio Grande Rift is reasonably well constrained to be 1° to 3° [*Hamilton*, 1981; *Cordell*, 1982; *Hamilton*, 1988]; therefore it is the

pre-Rio Grande Rift Colorado Plateau rotation which must have been greater than estimated by Hamilton. A degree-by-degree inquiry demonstrates that rotations of 6° to 12° of pre-Rio Grande Rift rotation and 1°–3° of subsequent rift-related rotation create the best statistical coincidence of coeval craton and CP paleopole pairs.

[59] Nevertheless, all rotations cause significant divergence of upper Morrison Formation paleopoles (Figure 6b). The statistically indistinguishable nature of the upper Morrison paleopoles prior to any rotation (Figure 6a) stands in stark contrast to all other CP-craton pole pairs (Figure 6c). Essentially no clockwise rotation of the upper Morrison pole around the Laramide Euler pole [*Hamilton*, 1988] is permissible at the 95% confidence level without causing statistical divergence.

[60] The convergence of the lower Morrison paleopoles and divergence of the upper Morrison poles due to rotating the Colorado Plateau counterclockwise suggests one of two things: (1) that the upper Morrison cratonic pole [*Steiner et al.*, 1994] contains some substantial error, or (2) that the Colorado Plateau experienced some additional tectonic displacement in mid-Morrison time. The first possibility seems highly unlikely, not because it is my data, but because the relative quality of the upper Morrison data, as judged by the number of samples exhibiting univectorial-to-origin demagnetization behavior and by the k value of the sample population. These factors make the cratonic upper Morrison paleopole as good or better than the cratonic lower Morrison pole. Tectonic problems can be ruled out as the beds are essentially flat lying and the same locality gave lower Morrison and Summerville paleopoles that are the same as those derived from dipping (35°)strata to the east.

[61] The second possibility, that there was independent rotation of the Plateau in mid-Morrison time, has supporting geologic data. Three robust geologic observations suggest or require a mid-Morrison hiatus: (1) the significant difference in paleopole positions between the lower and upper members of the formation, observed both on the Colorado Plateau and in cratonic New Mexico, cannot be explained without a time break; (2) the pole differences correspond to an abrupt change in formation lithology, from dominantly channel sandstone deposition to dominantly mudstone deposition, indicating a significant change in sea level or base level in mid-Morrison time; and (3) an onset of volcanism corresponds to the change in lithology and pole position. The differing indications of Colorado Plateau rotation between the upper and lower parts of the formation and the onset of nearby volcanism may suggest tectonism accompanying the hypothesized hiatus. Timing of these theorized mid-Morrison events is well constrained. Both magnetostratigraphic correlation of the Morrison Formation to the marine magnetic anomaly timescale [*Steiner et al.*, 1994] and radiometric dating within the Morrison Formation [*Kowallis et al.*, 1998] indicate that the hypothesized events occurred near 150 Ma [*Steiner*, 1998]; extrapolation of radiometric data suggest approximately 152 Ma.

[62] The concept of a mid-Morrison tectonic event has been difficult to support because of a relative paucity of evidence of tectonism at that time in western North America. However, since first proposed [*Steiner et al.*, 1994], increasing evidence for tectonic events near 150 Ma continues to appear. To name at few: the Josephine Ophiolite was obducted at about 150 Ma; deformation, regional metamorphism, and intrusion associated with the Nevadan orogeny occurred at approximately ~157–150 Ma [*Schweickert et al.*, 1984; *Wright and Fahan*, 1988]. Because plutons of 150–155 Ma age are considered to be the evidence of cessation of Nevadan orogenic events at 150 Ma [*Wright and Fahan*, 1988; *Thorman et al.*, 1991; *Foster et al.*, 1992; *Dunne and Walker*, 1993], a clockwise rotation of the Colorado Plateau might be a response to the previous orogenic events.

[63] Regardless of whether the Plateau rotated during Morrison time, no combination of Euler poles and rotations that restore the pre-late Morrison Colorado Plateau poles to their coeval cratonic counterparts with 95% confidence also allow the upper Morrison CP and craton poles to coincide at the same confidence level. A scenario of 5° rotation of the Colorado Plateau prior to Laramide orogenesis (Morrison time) and 3° rotation of the Plateau as a consequence of that orogeny brings coeval CP-craton paleopoles into statistical coincidence at 95% confidence and maintains statistical coincidence of the upper Morrison paleopoles at about a 90% confidence limit. Substantial changes in Euler pole to various positions around the perimeter of the CP can effect very similar but no better results. Clearly many more studies, particularly of the cratonic Late Morrison paleopole, are needed to resolve this conundrum.

3.5. Mid-Jurassic Apparent Polar Wander

[64] The Summerville paleopole of this study bears strongly on the debate over the true path of NA APW during the Jurassic [e.g., *Butler et al.*, 1992; *Van Fossen and Kent*, 1992]. The debate centers around the very appreciable difference between similar age Middle Jurassic paleopoles, the Moat Volcanics (~166 Ma) and Corral Canyon (172 Ma) paleopoles. Unfortunately, each one of those poles carries associated irresolvable uncertainties: igneous rock paleopoles including the Moat Volcanics possess a paleohorizontal uncertainty, whereas the Corral Canyon sites lie within a tectonic block which subsequently has been detached from the North American craton. Studies of other Basin and Range blocks in the geographic area have demonstrated that these blocks have been rotated relative to cratonic North America [e.g., *Hagstrum and Lipman*, 1991]. Thus without corroborating evidence, the Corral Canyon pole cannot be considered a reliable reflection of NA Jurassic APW. The paleohorizontal uncertainty of the Moat Volcanics paleopole is not removed by any of the arguments advanced by *Van Fossen and Kent* [1992], therefore that uncertainty remains.

[65] Numerous additional arguments further question the validity of the Moat Volcanics paleopole as a representative of North American APW in the Middle Jurassic. First, the Summerville Formation of this and earlier studies is only slightly younger than the Moat Volcanics extrusive event. Although not radiometrically determined, the age of the Summerville Formation is known through its interfingering

relationship with subjacent Middle Callovian marine deposits, thus the Summerville Formation is inferred to be Late Callovian [e.g., *Imlay*, 1980]. The timescale of *Gradstein et al.* [1995] indicates that the late Callovian spanned approximately 159 to 162 Ma; whereas the Jurassic timescale of *Palfry et al.* [2000] defines Late Callovian as approximately 157 to158 Ma. Hence, the Summerville paleopole is potentially 5 to 8 m.y. younger than that of the Moat Volcanics.

[66] An abundance of data provide very strong support for the paleopole position recorded by the Summerville, in contrast to the single Moat Volcanics determination. The statistically similar Summerville and lower Morrison Formations form a large block of paleomagnetic data; five paleopoles (from cratonic NA and a rotation-corrected Colorado Plateau) indicate the same paleopole location (Figure 6b). This large number of results constitutes very strong evidence that this paleopole is representative of North America in late Middle-earliest Late Jurassic time.

[67] It was argued [*Van Fossen and Kent*, 1992] that the Moat Volcanics pole likely reflected the paleoposition of North America of that time because a similar pole is predicted by other plates with Pangea; an updated similar analysis [*Courtillot et al.*, 1994] also came to the conclusion that the Pangean plate paleopoles could be use to assess the NA APW path in the Jurassic. However, stretching of the Pangean supercontinent crust already had occurred by Late Triassic time. Stretching and graben formation created rift basins around the North American margins, between and Europe, Africa, and South America (summarized by *Manspeizer* [1994]). North America finally broke from the Pangean supercontinent and seafloor began to form around 170 Ma [*Klitgord and Schouten*, 1986]. Therefore, it is unlikely that North America was rigidly attached to any of the other Pangean plates by the time the Moat Volcanics were formed and magnetized.

[68] Examination of the kinematics of the opening of the central Atlantic Ocean also provides evidence that the Moat Volcanics paleopole likely does not represent the position of NA with respect to the geomagnetic pole at 166 Ma. If it did, it should be possible to rotate reliable pre-rift North American paleopoles to the Moat Volcanics using the poles of rotation specified for early Atlantic Ocean opening [*Klitgord and Schouten*, 1986]. Such rotation was attempted here with the late Early Jurassic Kayenta paleopoles [*Bazard and Butler*, 1992; *Steiner and Helsley*, 1974] after correction for rotation of the Colorado Plateau. The more recent paleopole of the Kayenta is identical to Late Triassic paleopoles [*Bazard and Butler*, 1992], so the test applies to both the Early Jurassic and Late Triassic paleopoles. Neither pole assemblage can be restored to the Moat Volcanics pole position through the use of the rotation poles derived from the early opening of the central Atlantic Ocean.

[69] In addition, inclusion of the Moat Volcanics paleopole into the NA APW path would require an exceedingly large amount of North American plate motion. The acceptable NA paleopole older than the Moat Volcanics is that of the early Late Jurassic Kayenta Formation (corrected for the rotation of the Colorado Plateau). In the intervening approximately 22 m.y. between magnetization of the Kayenta and Moat Volcanics formations, North America would have had to travel sufficiently to create the 30° of arc-separation between the two; subsequently, an even larger additional motion would be necessary to generate the 44° of arc-separation between Moat Volcanics and Summerville paleopoles. The implied amounts of North American plate motion are far larger than can be accommodated by the slow spreading Atlantic Ocean (maximum spreading half-rate of 2 cm/yr).

4. Conclusions

[70] A new cratonic late Middle Jurassic paleopole, 148.2°E, 57.3°N (k = 31.0, α_{95} = 6.0), has been determined for North America from the late Callovian Summerville Formation of north central New Mexico.

[71] The new cratonic Summerville paleopole of this study forms the seventh coeval cratonic North American and Colorado Plateau paleopoles pairs. Despite the fact that the clockwise rotation of the Colorado Plateau consisted of multiple rotation events, the Plateau paleomagnetic poles can be restored to their cratonic counterparts by approximately 9 ± 3° of counterclockwise rotation around an Euler pole near the southeast edge of the Plateau.

[72] The late Middle Jurassic Summerville and early Late Jurassic lower Morrison Formation give statistically identical paleomagnetic poles; the observation is valid from both craton and Colorado Plateau paleopoles. This paleopole coincidence suggests that the stratigraphic J-5 unconformity encompasses little elapsed time, in agreement with the characteristics of the contact between the two formations.

[73] The abundance of Summerville-lower Morrison determinations of the same paleopole (5) presents a substantial database which argues for North American apparent polar wander to have tracked through that paleopole, thus validating the 60° latitudinal track in the existing debate over Jurassic NA APW. Including the Moat Volcanics paleomagnetic pole [*Van Fossen and Kent*, 1992] in the NA APW path requires excessive NA plate motion that is not supported by Atlantic seafloor spreading. Hence the North American APW path cannot include the Moat Volcanics paleomagnetic pole during mid-Jurassic time.

Appendix A: Critical Examination of *Molina-Garza et al.*'s [1998] Colorado Plateau Rotation Analysis

[74] That analysis of Colorado Plateau rotation employed a number of inappropriate data. Criticism of parts of the database used by *Molina-Garza et al.*'s [1998] follows.

1. Data from fragmental blocks at the edges of the major structural blocks (craton and CP) were included in this analysis. To obtain as precise a rotation value as possible, especially in the case of small rotations and small data sets, data from small blocks or deformed terrane between the major structural blocks under comparison must be excluded; these cannot be relied upon not to have rotated by amounts different from the main blocks (potentially in ball-bearing fashion). In seemingly inconsistent logic, *Molina-Garza et al.* [1998] correctly exclude data from blocks in the Basin

and Range structural province to the south of the Colorado Plateau, but nevertheless include Early and Late Triassic data from small blocks within the Rio Grande Rift, the eastern boundary of the CP. Their justification, the similarity of the paleopoles from some of these blocks to those of the CP, constitutes circular reasoning. Data also were included from the deformed region forming the northeastern boundary of the Plateau; the Maroon, Minturn, State Bridge (termed "Moenkopi" by the authors), Fountain, and Lykins Formations all lie in this zone of deformation and therefore should be viewed with suspicion.

2. *Molina-Garza et al.* [1998] do not consider the Rio Grande Rift to be the structural boundary between the Colorado Plateau and cratonic North America. They incorporate into their Colorado Plateau paleopole group, a paleopole that was derived from a locality lying east of the Rio Grande Rift, the Early Permian Abo paleopole from central New Mexico. They argue that, statistically, the pole lies closer to Colorado Plateau poles than craton poles, and thus assign it to the Colorado Plateau data bin [*Molina-Garza et al.*, 1998]. Four Early Permian paleopoles have been determined from cratonic North America. The Abo paleopole statistically lies no further from the group mean than any of the other three paleopoles; this topic was discussed in more detail in the text, within the Permian comparison of CP and NA paleopoles.

Regardless of the statistical affinities, the site from which the pole was derived is not tectonically part of the Colorado Plateau. Observations and analyses by *Kelley* [1979] and *Chamberlin and Anderson* [1989], among others, define the Rio Grande Rift as the structural boundary between the CP microplate and the North American plate; further support for this location of the boundary is demonstrated by geophysical data (particularly seismic and gravity data; see overview by *Keller et al.* [1998]) and by the demonstrated widening of the rift during Tertiary times [*Cordell*, 1982]. The prominence of surface geologic features (faulting, uplift and magmatism) all testify to the Rio Grande Rift as the structural boundary of consequence.

3. Incorporation of improperly demagnetized data: Nearly all red bed studies prior to about 1969 did not use thermal demagnetization. Examples of the data that *Molina-Garza et al.* [1998] should have excluded are the study by *McMahon and Strangway* [1968], which used only alternating field demagnetization, an inadequate treatment for red bed strata. The same is true of the studies of *Black* [1964], *Helsley* [1965], *Roy* [1966], and *Roy* [1969]. These data have not been properly demagnetized and should not be included in the analysis.

4. Use of duplicate data sets in their analysis [*Molina-Garza et al.*, 1998, Table 1]; this is obviously statistically invalid because it over weights certain results with respect to others. The Early Triassic Plateau Moenkopi paleopole of *Helsley* [1969] was derived from the same locality and samples as that of *Helsley and Steiner* [1974]; the 1974 analysis incorporated the 1969 data set. Moreover, the *Baag and Helsley* [1974] study is also from the very same locality, although that data set was derived from the subsurface. Similarly, *Helsley*'s [1971] study of the Cutler Formation was superceded by *Gose and Helsley* [1972]; again, the same locality and samples were used for the later analysis. A third such situation involves the Middle Triassic paleopole of *Molina-Garza et al.* [1996]; that paleopole is actually a mean of several studies, including the pole they listed just below that pole in their Table 1 (Middle Triassic paleopole of *Steiner and Lucas* [1992]), again constituting an invalid weighting of certain data. Finally, the upper and lower Pictou studies [*Symons*, 1990; *Pan and Symons*, 1993] superceded the study by *Black* [1964] in employing more thorough demagnetization.

5. The *Molina-Garza et al.* [1998] analysis uses plutons for which paleohorizontal is not known. Four such poles from northeastern NA were incorporated in the analysis (their Table 1). Furthermore, two of those poles were published as Middle Triassic, but were incorporated by *Molina-Garza et al.* [1998] into their Late Triassic (Carnian) bin.

6. Inclusion of the Newark Basin data: A comparison of solely cratonic Chinle Group paleopoles with those of the coeval Newark Supergroup has clearly shown that the Newark Basin is rotated with respect to cratonic North America [*Steiner and Lucas*, 2000]. Substantial databases underlie the poles of each of the stratigraphic groups; the Newark poles are observed to be rotated $9 \pm 3^\circ$ counterclockwise relative to Chinle poles of cratonic North America [*Steiner and Lucas*, 2000]. Although faunal evidence and magnetostratigraphy demonstrate that the two Late Triassic stratal groups are precisely coeval in age, their age equivalence is unimportant because both groups demonstrate that the location of the paleomagnetic pole changed little with respect to North America during the Late Triassic. Therefore, the Newark Basin paleopoles cannot be used to represent cratonic North America.

7. Use of a paleopole of unknown age: The large amount of APW during the time interval for which coeval CP and NA paleopoles exist requires that the age of all paleopoles used in an assessment of CP rotation be clearly known. *Molina-Garza et al.* [1998] use a pole derived from an overprint magnetization in the Dockum Group ("Dockum B"); no direct evidence for the age of that remanence exists. *Molina-Garza et al.* [1995] inferred an age by comparison of the cratonic remanence to Colorado Plateau paleopoles. No independent validation of that age is available, and moreover, the age was inferred by reference to Colorado Plateau paleopoles corrected for only 3° of Plateau rotation. Use of a cratonic paleopole to assess the amount of CP rotation when that paleopole's age is known only by reference to CP paleopoles constitutes a circular argument.

8. Mysterious poles: In their quest to operate on as large a database as possible, *Molina-Garza et al.* [1998] quote two paleopoles used in their analysis that were not computed in the original publications, nor were data with which to compute paleopoles provided in the original publications. No Early Triassic Moenkopi pole was computed by *Baag and Helsley* [1974], and no Early Permian Cutler pole was computed by *Reynolds et al.* [1985].

9. Exclusion of the cratonic lower Morrison pole: *Molina-Garza et al.* [1998] incorrectly state that the author stated that she considered the data unreliable; I did not.

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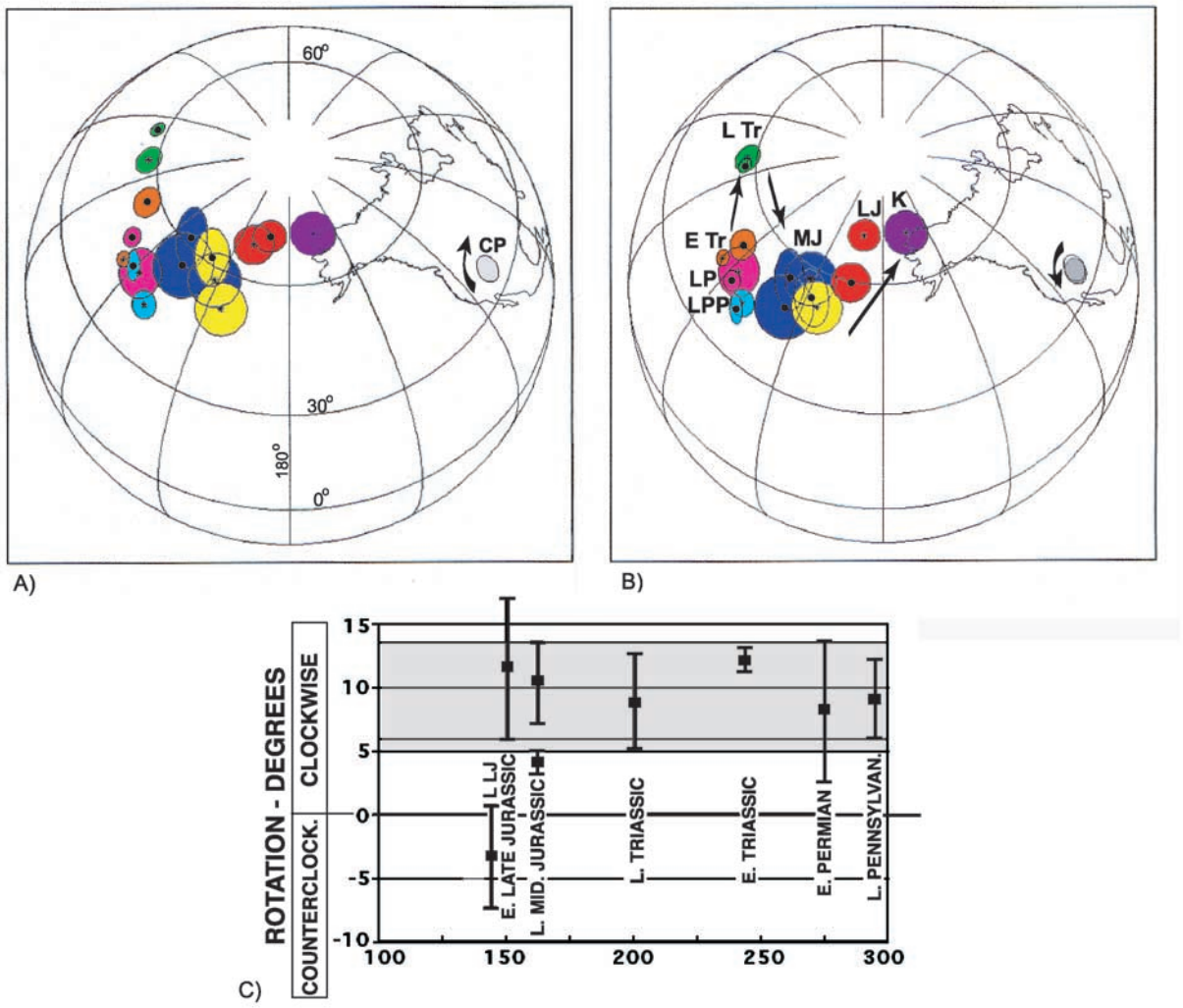


Figure 6. (a) Coeval paleopoles from the Colorado Plateau (solid dots) and stable NA craton (asterisks): Late Pennsylvanian (LPP, light blue); Early Permian (EP, rose pink); Early Triassic (ETR, orange); Late Triassic Chinle Group (LTR, green); Late Middle Jurassic, Callovian, Summerville (MJ, blue); early Late Jurassic, Oxfordian-lower Kimmeridgian lower Morrison Formation (yellow); and late, Late Jurassic, Kimmeridgian-Tithonian upper Morrison Formation (red). For reference, the averaged NA Late Cretaceous pole (Globerman and Irving, 1998) is shown in maroon. The Colorado Plateau is shaded gray. (b) Paleopoles of Figure 6a after the Colorado Plateau poles have been restored by a single 9° counterclockwise rotation around an Euler pole located at 34.6°N, 254.6°E. Arrows along the path APW path indicate the age progression of paleopoles. (c) The amount of rotation indicated by various Colorado Plateau paleopoles plotted againts paleopole age; clockwise (counterclockwise) rotation is positive (negative).